



UNIVERSITEIT VAN AMSTERDAM

Causality

NOTES BY

Hengyi Li

LECTURERS

Dr. P. D. Forré

Prof. dr. J. M. Mooij

INTRODUCTION. In this course, you will learn how to model causality mathematically, reason formally about cause, effect, and counterfactuals, predict the consequences of actions and analyse data to answer questions of a causal nature. We will use two different probabilistic frameworks for modelling causality: *causal Bayesian networks* and *structural causal models*. Topics addressed will be *causal modelling* (definition of Markov kernels, conditional independences, causal Bayesian networks, structural causal models, marginalisation, confounders, selection bias, feedback loops, causal graphs, interventions, Markov properties), *causal reasoning and estimation* (intervention variables, do-calculus, counterfactuals, covariate adjustment, back-door criterion, identifiability), and *causal discovery and estimation* (randomised controlled trials, instrumental variables, local causal discovery, Y-structures, the FCI algorithm).

Contents

1	Experimental Causal Discovery	1
1.1	Types of Correlations	1
1.2	Causal Effects in the Real World	2
1.3	Randomised Controlled Trials	3
2	Transition Probability Theory	5
2.1	Elementary Probability Theory	5
2.2	Measure Theoretic Probability	7
2.3	Transition Measures and Markov Kernels	11
2.4	Constructing Markov Kernels from Others	15
2.5	Conditional Independence	20
2.6	Separoid Axioms for Conditional Independence	24
3	Graph Theory	31
3.1	σ -Separation	34
3.2	d -Separation	37
3.3	Separoid Axioms for σ -/ d -Separation	38
4	Causal Bayesian Networks	42
4.1	Global Markov Property	43
4.2	Operations on Causal Bayesian Networks	44
4.3	Standard Forms of Causal Bayesian Networks	45
5	Identification of Causal Effects in CBNs	50
5.1	Do-Calculus	50
5.2	Adjustment Criteria and Formulae	52
6	Structural Causal Models	56
6.1	Solving the Structural Equations	56
6.2	Equivalence Notions	58
6.3	Interventions	59

6.4	Unique Solvability and Simple SCMs	60
6.5	Marginalisation	62
6.6	Acyclifications	63
7	Causal Domain Adaptation with SCMs	74
7.1	Do-Calculus for Simple SCMs	74
8	Counterfactuals with SCMs	75
8.1	Bounding Counterfactual Probabilities	75
9	Causal Relations and Confounding	79
10	Causal Discovery & Estimation with SCMs	81
11	Independence Testing	83
A	Final Exams	85
	References	90

Experimental Causal Discovery

SECTION 1.1

| Types of Correlations

EXAMPLE 1.1.

Remark. Correlation does not imply causation because there are other possible correlation inducing scenarios. Also, correlation is symmetric, causation is asymmetric.

DEFINITION 1.2. (i) *Deterministic systems:* If a minimal intervention that sets the value of X changes the possible values of Y , we say that X causes Y .

(ii) *Stochastic systems:* If a minimal intervention that sets the value of X changes the probability distribution of Y , we say that X causes Y .

SECTION 1.2

| Causal Effects in the Real World

EXAMPLE 1.3. Consider an old type of thermometer (T) with a needle that can either point to higher temperatures (up) or to lower temperatures (down). We also consider the state of the sun (S), which can either be up (u) or down (d). We then observe that T correlates with S . For simplicity, we assume a one-to-one relationship:

T	S
u	u
d	d

The conditional distribution $\mathbb{P}(S \mid T)$ then looks like this:

$$\begin{aligned}\mathbb{P}(S = u \mid T = u) &= 1, \\ \mathbb{P}(S = u \mid T = d) &= 0, \\ \mathbb{P}(S = d \mid T = u) &= 0, \\ \mathbb{P}(S = d \mid T = d) &= 1.\end{aligned}$$

It makes clear that there is a difference between:

- (i) *Observing* the positions of the thermometer needle T and the sun S , resulting in an *observational* data set, leading to an estimate of $\mathbb{P}(S \mid T)$.
- (ii) *Interacting* with the thermometer needle T and getting the sun's response S , resulting in an *interventional* data set, leading to estimates for $\mathbb{P}(S \mid \text{do}(T))$:

$$\begin{aligned}\mathbb{P}(S = u \mid \text{do}(T = u)) &= 0.5, \\ \mathbb{P}(S = u \mid \text{do}(T = d)) &= 0.5, \\ \mathbb{P}(S = d \mid \text{do}(T = u)) &= 0.5, \\ \mathbb{P}(S = d \mid \text{do}(T = d)) &= 0.5.\end{aligned}$$

DEFINITION 1.4.

We (informally) say that a variable X has a *causal effect* on another variable Y if forcing X to take on a value x , the distribution of Y explicitly depends on x , that is:

$$\exists x \in \mathcal{X} : \mathbb{P}(Y \mid \text{do}(X = x)) \neq \mathbb{P}(Y).$$

This is different from

$$\exists x \in \mathcal{X} : \mathbb{P}(Y \mid X = x) \neq \mathbb{P}(Y)$$

which just uses the conditional distributions instead of the interventional distributions.

Remark. (i) The *do*-operators are not operators on the observational distribution $\mathbb{P}(X, Y)$ or $\mathbb{P}(Y \mid X)$, etc., or on the corresponding observational data sets. They reflect actions/interventions in the real world leading to different distributions and corresponding data sets.

- (ii) There are usually many possible intervention values and targets one can think of, leading to many different interventional distributions and data sets. One may think of the observational distribution as a special case of an interventional distribution (where we intervene by doing nothing).

SECTION 1.3

| Randomised Controlled Trials

EXAMPLE 1.5 (Randomised Controlled Trials). Assume we want to know if *treatment* variable X has a causal effect on *outcome* variable Y , i.e., we want to estimate the deviation between $\mathbb{P}(Y \mid \text{do}(X = x_0))$ and $\mathbb{P}(Y \mid \text{do}(X = x_1))$. For this we have test subjects $w_1, \dots, w_N$. A *randomised controlled trial* then follows the following steps:

- (i) Split the population of test subjects into 2 groups (*test group* C_1 vs. *control group* C_0) by random lot (or fair coin flips).
- (ii) Give every test subject $w_n \in C_1$ from test group the treatment x_1 and the ones $w_n \in C_0$ from control group the control treatment x_0 .
- (iii) Measure the outcome y_n for each test subject w_n and estimate the deviation:

$$D := d(\mathbb{P}(Y \mid \text{do}(X = x_0)), \mathbb{P}(Y \mid \text{do}(X = x_1))).$$

- (iv) Do a statistical test if the deviation D is significantly different from 0.
- (v) If it is significantly different from 0 we can conclude a *causal effect* of X on Y , otherwise, the test does not establish a causal effect.

Remark.

THEOREM 1.6.

EXAMPLE 1.7.

THEOREM 1.8.

For discrete X, Y, Z , even with unobserved confounding,

$$\mathbb{P}(X = x, Y = y) \leq \mathbb{P}(Y(x) = y) \leq \mathbb{P}(X = x, Y = y) + 1 - \mathbb{P}(X = x).$$

Proof. Using consistency $Y = Y(X)$ and elementary probability theory:

$$\begin{aligned} \mathbb{P}(X = x, Y = y) &= \mathbb{P}(X = x, Y(x) = y) \\ &\leq \mathbb{P}(Y(x) = y) \\ &= \mathbb{P}(X = x, Y(x) = y) + \mathbb{P}(X \neq x, Y(x) = y) \\ &\leq \mathbb{P}(X = x, Y = y) + \mathbb{P}(X \neq x) \\ &= \mathbb{P}(X = x, Y = y) + 1 - \mathbb{P}(X = x). \end{aligned}$$

□

| Exercises

EXERCISE 1.1 (Correlation and Causal Hypotheses). For each observation below, give three possible causal explanations, allowing additional variables. Describe each explanation in words and draw its causal graph.

- (a) A half-dose vaccine immunises 90% of patients in country A, whereas a full dose immunises 60% in country B.
- (b) Group A comprises 10% of the population and group B 90%, but 90% of those arrested for criminal offences belong to group A.
- (c) High-school graduation grades strongly predict later career success.

EXERCISE 1.2 (Causal and Non-Causal Predictions). Choose three systems. For each, give one causal prediction and one non-causal prediction that could be made from data on that system.

EXERCISE 1.3 (Surrogate Endpoints). Let T denote Aduhelm treatment, β amyloid beta plaque levels, and O the clinical outcome of an Alzheimer's patient. The accelerated-approval argument relies on randomised trials showing plaque reduction, while critics report insufficient evidence that this reduction slows cognitive and functional decline.

- (a) Specify the assumed causal relations E , where $(A, B) \in E$ means that A causes B .
- (b) Propose an alternative graph explaining dependence between plaque levels and disease, and a treatment effect on plaque, without an effect of treatment on clinical outcome.

SOLUTION. Let T denote treatment with Aduhelm, β the amyloid beta plaque level, and O the clinical outcome. The surrogate-endpoint argument assumes

$$T \longrightarrow \beta \longrightarrow O.$$

A randomised trial demonstrating an effect of T on β does not by itself establish an effect on O . An alternative graph is

$$T \longrightarrow \beta \longleftarrow F \longrightarrow O,$$

where F represents a common cause of plaque levels and clinical outcomes. This graph permits dependence between β and O and a treatment effect on β , without a directed path from T to O . $\square$

EXERCISE 1.4 (Assumptions of Randomised Controlled Trials). For treatment C and outcome X , RCT reasoning assumes (i) X does not cause C , and (ii) C, X are not confounded because treatment assignment is independent of latent factors influencing outcome. Give one situation violating each assumption.

Transition Probability Theory

SECTION 2.1

| Elementary Probability Theory

EXAMPLE 2.1 (A Biased Die). You are allowed to roll a biased die with 6 sides. If you roll a 5 or 6 you win a *car*, a 4 gives you a *mug* and 1,2,3 wins you an apple *pie*. In this case the sample space is $\mathcal{W} := \{1, 2, 3, 4, 5, 6\}$ and the die introduces a probability distribution $\mathbb{P}$ on $\mathcal{W}$. Since the die is biased, we have to specify each of the probability masses to throw those numbers separately:

$$\mathbb{P}(1) = 0.5, \quad \mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = 0.1, \quad \mathbb{P}(5) = 0.15, \quad \mathbb{P}(6) = 0.05.$$

We are now interested in the probabilities of the events of winning those 3 different prizes. For this we consider the ‘prize’ space: $\mathcal{Z} := \{\text{pie}, \text{mug}, \text{car}\}$. We can then formalise the outcome via the map F :

$$\begin{aligned} F : \mathcal{W} &\rightarrow \mathcal{Z}, \\ 1, 2, 3 &\mapsto \text{pie}, \\ 4 &\mapsto \text{mug}, \\ 5, 6 &\mapsto \text{car}. \end{aligned}$$

To compute the probability of winning each of the prizes we need to ‘push’ the probability distribution $\mathbb{P}$, which lives on the space $\mathcal{W}$, to the space $\mathcal{Z}$. We can do this as follows:

$$\begin{aligned} \mathbb{P}(F = \text{pie}) &= \mathbb{P}\left(F^{-1}(\{\text{pie}\})\right) = \mathbb{P}(\{1, 2, 3\}) = \mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) = 0.7, \\ \mathbb{P}(F = \text{mug}) &= \mathbb{P}\left(F^{-1}(\{\text{mug}\})\right) = \mathbb{P}(\{4\}) = \mathbb{P}(4) = 0.1, \\ \mathbb{P}(F = \text{car}) &= \mathbb{P}\left(F^{-1}(\{\text{car}\})\right) = \mathbb{P}(\{5, 6\}) = \mathbb{P}(5) + \mathbb{P}(6) = 0.2, \end{aligned}$$

where $F^{-1}(C) := \{w \in \mathcal{W} \mid F(w) \in C\}$ is the *pre-image* of $C \subseteq \mathcal{Z}$.

Remark. (i) We call the tuple $(\mathcal{W}, \mathbb{P})$ a *probability space*. It is important to note that $\mathbb{P}$ was defined on $\mathcal{W}$, not $\mathcal{Z}$.

(ii) We call the map F a *random variable*, which is really nothing else than a map from a probability space to another space.

(iii) Events are modelled by subsets $B \subseteq \mathcal{W}$, not just by single elements $w \in \mathcal{W}$. For example consider the event that you don’t win a car. This event can’t be represented by a single element in $\mathcal{W}$ or $\mathcal{Z}$.

- (iv) In this example we can compute the probability of an event by additivity of $\mathbb{P}$ and the use of the probability mass function, via $\mathbb{P}(B) = \sum_{w \in B} p(w)$.
- (v) The distribution of the prizes, i.e., the distribution of random variable F , assigns probabilities to events $C \subseteq \mathcal{Z}$ and can be computed using the pre-image of F via $\mathbb{P}(F \in C) = \mathbb{P}(F^{-1}(C))$, where the latter is now an event $F^{-1}(C) \subseteq \mathcal{W}$, which we already know how to deal with.

EXAMPLE 2.2 (Normal Distribution). Consider the standard normal distribution $\mathcal{N}(0, 1)$ on $\mathbb{R}$, which is specified by the probability density function:

$$p(w) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}w^2\right).$$

The probability of an event $A \subseteq \mathbb{R}$ is then given by:

$$\mathbb{P}(A) = \int_A p(w) dw$$

in case A can be integrated over (i.e., if it is not a too pathological set). For instance, if $A = [a, b] \cup [c, d]$ with $a \leq b < c \leq d$ we get:

$$\mathbb{P}(A) = \int_a^b p(w) dw + \int_c^d p(w) dw.$$

Note that, even though $p(w) > 0$ for every $w \in \mathbb{R}$, we have:

$$\mathbb{P}(\{x\}) = 0, \quad \forall x \in \mathbb{R}.$$

Now consider the random variable $F : \mathbb{R} \rightarrow \mathbb{R}$ with $F(w) = \sin(w)$. It is not immediately clear how to define the probability distribution of F when only working with probability densities. It is even more difficult to derive the probability density for F in this setting.

- Remark.*
- (i) Many probability distributions can be represented either by probability mass functions (discrete case), $w \mapsto p(w)$, or probability density functions (absolutely continuous case), $w \mapsto p(w)$.
 - (ii) We have problems defining probability distributions of random variables for absolutely continuous distributions when we are only allowed to work with probability densities.
 - (iii) Measure theory is the framework that directly works with subsets $A \subseteq \mathcal{W}$, in contrast to elements $w \in \mathcal{W}$, and provides a unifying language that encompasses both special cases.

SECTION 2.2

| Measure Theoretic Probability

Measurable Spaces and Maps

DEFINITION 2.3 (σ -algebras).

Let $\mathcal{W}$ be a set. A (non-empty) set $\mathcal{B} \subseteq 2^{\mathcal{W}}$ of subsets $A \subseteq \mathcal{W}$ is called a σ -algebra on $\mathcal{W}$ if it satisfies the following rules:

- (i) *empty set*: $\emptyset \in \mathcal{B}$.
- (ii) *complement*: If $A \in \mathcal{B}$ then also $A^c := \mathcal{W} \setminus A \in \mathcal{B}$.
- (iii) *countable union*: If $A_n \in \mathcal{B}$ for all $n \in \mathbb{N}$ then also: $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{B}$.

DEFINITION 2.4 (Measurable Spaces).

A tuple $(\mathcal{W}, \mathcal{B})$ of a set $\mathcal{W}$ and a σ -algebra $\mathcal{B}$ on $\mathcal{W}$ is called a *measurable space*.

Remark. By abuse of notation we often just call $\mathcal{W}$ a measurable space by implicitly assuming that it is endowed with a fixed σ -algebra, which we will indicate by $\mathcal{B}_{\mathcal{W}}$ or $\mathcal{B}(\mathcal{W})$ if needed. We will also just call a subset $A \subseteq \mathcal{W}$ measurable when we actually mean that $A \in \mathcal{B}_{\mathcal{W}}$.

EXAMPLE 2.5 (Measurable Spaces). (i) Finite spaces:

- (ii) Countably infinite spaces:
- (iii) Metric spaces:

For most of the notes we will restrict to well-behaved measurable spaces, namely *standard measurable spaces*. The key point is that they all behave like the space $[0, 1]$, or $\mathbb{R}$, with its Borel- σ -algebra. So (almost) all results for $[0, 1]$ immediately translate to standard measurable spaces.

DEFINITION 2.6 (Standard Measurable Spaces).

A measurable space $(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$ is called *standard measurable space* (or *standard Borel space*) if it is measurably isomorphic¹ to either:

- (i) A finite measurable space $\{1, \dots, M\}$ for some $M \in \mathbb{N}$ endowed with the power set σ -algebra $2^{\{1, \dots, M\}}$.
- (ii) The countably infinite space $\mathbb{N}$ endowed with the power set σ -algebra $2^{\mathbb{N}}$.
- (iii) The unit interval $[0, 1]$ endowed with its Borel σ -algebra:

$$\mathcal{B}_{[0,1]} = \sigma(\{[a, b] \mid a, b \in [0, 1] \cap \mathbb{Q}, a \leq b\}),$$

which is the smallest σ -algebra that contains all intervals.

The following theorem shows that (almost) all spaces we encounter in practice are actually standard measurable spaces, justifying our focus on standard measurable spaces for most of the notes.

¹ *Measurably isomorphic* means that there is a measurable map from one space to the other that has a measurable inverse.

THEOREM 2.7 (Kuratowski et al.).

Every Borel subset of any complete metric space $(\mathcal{W}, d)$ that has a countable dense subset is a standard measurable space in its Borel σ -algebra.

EXAMPLE 2.8 (Standard Measurable Spaces). $\mathbb{R}, \mathbb{R}^D, \mathbb{Q}, \mathbb{Z}, \mathbb{N}, \{1, \dots, M\}, [0, 1]$, topological manifolds, countable CW-complexes, every Borel set of any separable complete metric space, etc.

DEFINITION 2.9 (Measurable Maps).

Let $(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$ and $(\mathcal{Z}, \mathcal{B}_{\mathcal{Z}})$ be two measurable spaces and $f : \mathcal{W} \rightarrow \mathcal{Z}$ be a map. We call f a $\mathcal{B}_{\mathcal{W}}$ - $\mathcal{B}_{\mathcal{Z}}$ -measurable map (or just measurable for short) if for all $B \in \mathcal{B}_{\mathcal{Z}}$ the pre-image² $f^{-1}(B)$ is an element of $\mathcal{B}_{\mathcal{W}}$. In formulas:

$$\forall B \in \mathcal{B}_{\mathcal{Z}} : f^{-1}(B) \in \mathcal{B}_{\mathcal{W}}.$$

EXAMPLE 2.10.

Finite and Probability Measures

DEFINITION 2.11 (Measures).

Let $(\mathcal{W}, \mathcal{B})$ be a measurable space. A measure μ on $(\mathcal{W}, \mathcal{B})$ is a map:

$$\mu : \mathcal{B} \rightarrow \mathbb{R} \cup \{\infty\}, \quad D \mapsto \mu(D)$$

such that:

- (i) non-negative: $\forall A \in \mathcal{B} : \mu(A) \in [0, \infty]$.
- (ii) empty set: $\mu(\emptyset) = 0$.
- (iii) countably additive (or σ -additive): for all sequences $A_n \in \mathcal{B}, n \in \mathbb{N}$, with $A_i \cap A_j = \emptyset$ for all $i \neq j$, we have:

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n).$$

Remark. If this holds, we call the triple $(\mathcal{W}, \mathcal{B}, \mu)$ measure space. The value ∞ is explicitly allowed.

DEFINITION 2.12.

A measure μ on $(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$ is called:

- probability measure if $\mu(\mathcal{W}) = 1$.
- finite measure if $\mu(\mathcal{W}) < \infty$.
- σ -finite measure if

Clearly, every probability measure is finite, and, every finite measure is σ -finite.

Remark. If μ is a probability measure then the triple $(\mathcal{W}, \mathcal{B}, \mu)$ is called probability space.

EXAMPLE 2.13. (i) Dirac measures:

² The definition of pre-image: $f^{-1}(B) := \{w \in \mathcal{W} \mid f(w) \in B\}$.

- (ii) *Discrete measures:*
- (iii) *The Lebesgue measure:*

DEFINITION 2.14 (Push-Forward Measure).

Let $X : (\mathcal{W}, \mathcal{B}_{\mathcal{W}}) \rightarrow (\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ be measurable and μ a measure on $(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$. Then we define the *push-forward measure* (or *image measure*) of μ via:

$$(X_*\mu)(A) := \mu_X(A) := \mu(X \in A) := \mu(X^{-1}(A))$$

for all $A \in \mathcal{B}_{\mathcal{X}}$. If μ is a probability distribution then the push-forward measure $\mu(X)$ is also called the *(distributional) law* of X .

DEFINITION 2.15 (Random Variable).

A measurable mapping:

$$X : (\mathcal{W}, \mathcal{B}_{\mathcal{W}}, \mathbb{P}) \rightarrow (\mathcal{X}, \mathcal{B}_{\mathcal{X}})$$

that starts from a probability space is also called a *random variable*.

Remark. The main point is that the map X comes with its own distribution P^X . We often just say: "Let X be a random variable with distribution $P^X = \dots$ ", where P^X is then specified, e.g. to be a Gaussian or a categorical distribution, etc.

NOTATION 2.16 (Measure Integral).

We abbreviate the *measure integral* of a measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$ w.r.t. measure μ on $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ as:

$$\int f d\mu = \int f(x) d\mu(x) = \int f(x) \mu(dx).$$

If $\mathbb{P}$ is a probability measure on $\mathcal{X} = \mathbb{R}^D$ that is either discrete or absolutely continuous we have:

$$\int f(x) \mathbb{P}(dx) = \begin{cases} \sum_{x \in \mathcal{X}} f(x) \cdot p(x), & \text{if } P \text{ is discrete} \\ \int_{\mathcal{X}} f(x) \cdot p(x) dx, & \text{if } P \text{ is absolutely continuous} \end{cases},$$

where p either denotes the probability mass function or the probability density, respectively.

DEFINITION 2.17 (Expectation Value).

Let $(\mathcal{W}, \mathcal{B}, P)$ be a probability space and $X : \mathcal{W} \rightarrow \mathbb{R}$ be a measurable function with well-defined integral. Then its *expectation value* (w.r.t. $\mathbb{P}$) is defined to be:

$$\mathbb{E}[X] := \int X d\mathbb{P}.$$

DEFINITION 2.18.

The set of all probability measures on $(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$ is denoted by $\mathcal{P}(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$, and the set of all finite measures by $\mathcal{M}(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$, or $\mathcal{P}(\mathcal{W})$ and $\mathcal{M}(\mathcal{W})$, respectively for short.

DEFINITION 2.19 (Evaluation Maps).

For $B \in \mathcal{B}_{\mathcal{W}}$ we consider the *evaluation map*:

$$ev_B : \mathcal{M}(\mathcal{W}) \rightarrow \mathbb{R}_{\geq 0}, \quad \mu \mapsto ev_B(\mu) := \mu(B).$$

We then endow $\mathcal{M}(\mathcal{W})$, and $\mathcal{P}(\mathcal{W})$, respectively, with the smallest σ -algebra $\mathcal{B}$ such that all evaluation maps ev_B are $\mathcal{B}$ - $\mathcal{B}_{\mathbb{R}_{\geq 0}}$ -measurable, where $\mathcal{B}_{\mathbb{R}_{\geq 0}}$ is the Borel- σ -algebra of $\mathbb{R}_{\geq 0}$, i.e.,

$$\mathcal{B}_{\mathcal{M}(\mathcal{W})} := \sigma\left(\{ev_B^{-1}((r, \infty)) \mid B \in \mathcal{B}_{\mathcal{W}}, r \in \mathbb{R}_{\geq 0}\}\right).$$

Remark. The above definition implies that for measurable spaces $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$, $(\mathcal{Y}, \mathcal{B}_{\mathcal{Y}})$, a map:

$$K : \mathcal{X} \rightarrow \mathcal{M}(\mathcal{Y}),$$

is $\mathcal{B}_{\mathcal{X}}$ - $\mathcal{B}_{\mathcal{M}(\mathcal{Y})}$ -measurable if and only if for all $B \in \mathcal{B}_{\mathcal{Y}}$ the composition:

$$ev_B \circ K : \mathcal{X} \rightarrow \mathcal{M}(\mathcal{Y}) \rightarrow \mathbb{R}_{\geq 0},$$

is $\mathcal{B}_{\mathcal{X}}$ - $\mathcal{B}_{\mathbb{R}_{\geq 0}}$ -measurable. Similarly, for $\mathcal{P}(\mathcal{Y})$.

THEOREM 2.20.

If $(\mathcal{W}, \mathcal{B}_{\mathcal{W}})$ is a standard measurable space then also $\mathcal{M}(\mathcal{W})$ and $\mathcal{P}(\mathcal{W})$ are standard measurable spaces (in their usual σ -algebra).

EXAMPLE 2.21. $\mathcal{P}(\mathbb{R}^d)$ is a standard measurable space.

SECTION 2.3

| Transition Measures and Markov Kernels

EXAMPLE 2.22 (Probabilistic Map). If we consider a deterministic measurable map $f : \mathcal{T} \rightarrow \mathcal{W}$ then f assigns to each point $t \in \mathcal{T}$ exactly one point $w = f(t) \in \mathcal{W}$. Sometimes we rather want to model a *probabilistic map*, i.e., an assignment that can be random or comes with some uncertainties but still changes depending on the input t . The notion of *Markov kernels* formalises this. A Markov kernel K from $\mathcal{T}$ to $\mathcal{W}$ can be considered a measurable map from $\mathcal{T}$ to the space of probability measures $\mathcal{P}(\mathcal{W})$ of $\mathcal{W}$:

$$\mathcal{T} \rightarrow \mathcal{P}(\mathcal{W}).$$

It assigns to each $t \in \mathcal{T}$ a probability distribution over $\mathcal{W}$, which then assigns to each measurable subset $D \subseteq \mathcal{W}$ a probability value in $[0, 1]$.

- EXAMPLE 2.23 (Markov Kernels). (i) Any statistical model (i.e., family of model distributions) $\{p_\theta \mid \theta \in \mathcal{F}\}$, can be considered a Markov kernel, which we write $P(X \mid \Theta)$.
(ii) Any conditional distribution $P(Y \mid X)$ can be considered a Markov kernel.
(iii) A neural network with Softmax output for classification with input $x \in \mathcal{X}$, output $y \in \mathcal{Y}$ and weights $w \in \mathcal{W}$ can be seen as a Markov kernel $P(Y \mid X, W)$.
(iv) Normal distribution as Markov kernel:

$$P(X \in A \mid T = t) = \int_A \mathcal{N}(x \mid t, \Sigma) dx.$$

DEFINITION 2.24.

Let $\mathcal{T}$ and $\mathcal{W}$ be measurable spaces, $\mathcal{P}(\mathcal{W})$ and $\mathcal{M}(\mathcal{W})$ the spaces of probability measures and finite measures, respectively, on $\mathcal{W}$.

- (i) A (*finite*³) *transition measure* from $\mathcal{T}$ to $\mathcal{W}$ is a measurable map:

$$K : \mathcal{T} \rightarrow \mathcal{M}(\mathcal{W}),$$

from $\mathcal{T}$ to the space of finite measures of $\mathcal{W}$.

- (ii) A *transition probability* or *Markov kernel* from $\mathcal{T}$ to $\mathcal{W}$ is a measurable map:

$$K : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{W}),$$

from $\mathcal{T}$ to the space of probability measures of $\mathcal{W}$.

NOTATION 2.25. (i) We often use suggestive notations as follows for finite transition measures and Markov kernels:

$$K(W \mid T) : \mathcal{T} \rightarrow \mathcal{M}(\mathcal{W}), \quad t \mapsto K(W \mid T = t),$$

where for every fixed $t \in \mathcal{T}$ the following map:

$$K(W \mid T = t) : \mathcal{B}_{\mathcal{W}} \rightarrow \mathbb{R}_{\geq 0}, \quad D \mapsto K(W \in D \mid T = t)$$

is a finite measure, or probability measure, respectively.

³ In this course we will only discuss finite transition measures and just drop the word “finite” for simplicity in the following.

(ii) For fixed $D \in \mathcal{B}_{\mathcal{W}}$ we then use the following notation for the following measurable map:

$$K(W \in D | T) : \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}, \quad t \mapsto K(W \in D | T = t).$$

(iii) Since $K(W | T)$ takes the argument $t \in \mathcal{T}$ first, but then also $D \in \mathcal{B}_{\mathcal{W}}$ as a second argument we can also identify $K(W | T)$ with the following two-argument map, which we denote with the same symbols:

$$K(W | T) : \mathcal{B}_{\mathcal{W}} \times \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}, \quad (D, t) \mapsto K(W \in D | T = t).$$

(iv) For Markov kernels $K(W | T)$ we will most of the time use the dashed arrow to $\mathcal{W}$ (instead of a usual arrow to $\mathcal{P}(\mathcal{W})$) to indicate the Markov kernel as follows:

$$K(W | T) : \mathcal{T} \dashrightarrow \mathcal{W}, \quad (D, t) \mapsto K(W \in D | T = t).$$

LEMMA 2.26.

There is a one-to-one correspondence between the following constructions:

(i) a Markov kernel, i.e., a measurable map:

$$K(W | T) : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{W}), \quad t \mapsto (D \mapsto K(W \in D | T = t))$$

(ii) a two-argument function:

$$\tilde{K}(W | T) : \mathcal{B}_{\mathcal{W}} \times \mathcal{T} \rightarrow [0, 1], \quad (D, t) \mapsto \tilde{K}(W \in D | T = t)$$

such that:

– for each $t \in \mathcal{T}$ the map:

$$\mathcal{B}_{\mathcal{W}} \rightarrow [0, 1], \quad D \mapsto \tilde{K}(W \in D | T = t)$$

is a probability measure, and

– for each $D \in \mathcal{B}_{\mathcal{W}}$ the map:

$$\tilde{K}(W \in D | T) : \mathcal{T} \rightarrow [0, 1], \quad t \mapsto \tilde{K}(W \in D | T = t)$$

is $\mathcal{B}_{\mathcal{T}}\text{-}\mathcal{B}_{[0,1]}$ -measurable and vice versa.

Proof. If K is measurable, then $K(t)$ is a probability measure and

$$\tilde{K}(D, \cdot) = ev_D \circ K$$

is measurable because each evaluation map ev_D is measurable. Conversely, define $K(t)(D) = \tilde{K}(D, t)$. The first condition ensures $K(t) \in \mathcal{P}(\mathcal{W})$. The evaluation σ -algebra is generated by $ev_D^{-1}(B)$, and

$$K^{-1}(ev_D^{-1}(B)) = \tilde{K}(D, \cdot)^{-1}(B) \in \mathcal{B}_{\mathcal{T}}.$$

Thus K is measurable. □

EXAMPLE 2.27 (Special Cases of Markov Kernels). (i) *Normal distribution.*

(ii) *Exponential families as finite transition measures.*

(iii) *Mass functions.* Consider a Markov kernel:

$$K(W | T) : \mathcal{T} \dashrightarrow \mathcal{W}, \quad (D, t) \mapsto K(W \in D | T = t),$$

where both $\mathcal{W} = \{w_1, \dots, w_M\}$ and $\mathcal{T} = \{t_1, \dots, t_K\}$ are finite discrete spaces. Then we can define the mass function k via:

$$k(w_i | t_j) := K(W \in \{w_i\} | T = t_j),$$

and the matrix $\tilde{K} := (k(w_i | t_j))_{i,j}$. Then the matrix $\tilde{K}$ is a stochastic matrix, i.e., it has non-negative entries and each of its columns sums to 1. $\tilde{K}$ then fully determines the Markov kernel K . So in the (finite) discrete case a Markov kernel is basically nothing else than a stochastic matrix filled with the transition probabilities.

- (iv) *Markov kernels generalise probability distributions.*
- (v) *Markov kernels generalise deterministic maps.* Consider a measurable mapping $f : \mathcal{T} \rightarrow \mathcal{W}$. Then we can turn f into a Markov kernel δ_f via:

$$\delta_f : \mathcal{T} \rightarrow \mathcal{W}, \quad (D, t) \mapsto \delta_f(D | t) := \mathbb{1}_D(f(t)),$$

which puts 100% probability mass onto the function value $f(t)$ for given $t \in \mathcal{T}$.

DEFINITION 2.28 (Alternative Definition).

A *transition measure* from measurable space $\mathcal{T}$ to $\mathcal{W}$ can be characterised as a map:

$$K(W | T) : \mathcal{B}_{\mathcal{W}} \times \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}$$

such that

- (i) for all $t \in \mathcal{T}$ the mapping:

$$K(W | T = t) : \mathcal{B}_{\mathcal{W}} \rightarrow \mathbb{R}_{\geq 0}, \quad B \mapsto K(W \in B | T = t)$$

is a finite/probability measure;

- (ii) for all $D \in \mathcal{B}_{\mathcal{W}}$ the mapping:

$$K(W \in D | T) : \mathcal{T} \rightarrow \mathcal{M}(\mathcal{W}) \xrightarrow{\text{ev}_D} \mathbb{R}_{\geq 0}, \quad t \mapsto K(W \in D | T = t)$$

is $\mathcal{B}_{\mathcal{T}}\text{-}\mathcal{B}_{\mathbb{R}_{\geq 0}}$ -measurable.

DEFINITION 2.29 (Transition Probability Space).

Consider measurable spaces $\mathcal{T}$ and $\mathcal{W}$ and a Markov kernel:

$$K(W | T) : \mathcal{T} \rightarrow \mathcal{W}, \quad (D, t) \mapsto K(W \in D | T = t).$$

Then we call the tuple $(\mathcal{W} \times \mathcal{T}, K(W | T))$ a *transition probability space*.

DEFINITION 2.30 (Null Sets).

Let $(\mathcal{W} \times \mathcal{T}, K(W | T))$ be a transition probability space.

DEFINITION 2.31 (Conditional Random Variables).

A measurable map:

$$X : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{X}, \quad (w, t) \mapsto X(w, t)$$

starting from a transition probability space $(\mathcal{W} \times \mathcal{T}, K(W | T))$ is called *conditional random variable*.

Remark. It generalises the notion of random variables and can be considered a family of random variables (measurably) parameterised by $t \in \mathcal{T}$.

EXAMPLE 2.32 (Normal Distribution).

SECTION 2.4

| Constructing Markov Kernels from Others

DEFINITION 2.33 (Composition of Markov Kernels).

Consider two Markov kernels:

$$Q(Z | Y, W, T) : \mathcal{Y} \times \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{Z}, \quad K(W, U | T, X) : \mathcal{T} \times \mathcal{X} \rightarrow \mathcal{W} \times \mathcal{U}.$$

Then we define their composition:

$$Q(Z | Y, W, T) \circ K(W, U | T, X) : \mathcal{Y} \times \mathcal{T} \times \mathcal{X} \rightarrow \mathcal{Z},$$

using measurable sets $C \subseteq \mathcal{Z}$ via:

$$(C, (y, t, x)) \mapsto \int Q(Z \in C | Y = y, W = w, T = t) K(W \in dw | T = t, X = x).$$

EXAMPLE 2.34. (i) *Composition of discrete Markov kernels.* Assume that all the spaces are discrete/finite and let

$$P(Z | T) := Q(Z | W) \circ K(W | T)$$

be the composition of Markov kernels. Let p, q, k denote the corresponding mass functions. Then we get:

$$p(z_i | t_j) = \sum_k q(z_i | w_k) \cdot k(w_k | t_j).$$

If $\tilde{P}, \tilde{Q}, \tilde{K}$ are the corresponding stochastic matrices then we have that:

$$\tilde{P} = \tilde{Q}\tilde{K},$$

is just the usual matrix product. So in this case the composition of Markov kernels corresponds to matrix multiplication.

(ii) *Composition of deterministic Markov kernels.* Consider measurable maps:

$$X : \mathcal{T} \rightarrow \mathcal{X}, \quad Z : \mathcal{X} \rightarrow \mathcal{Z},$$

and their composition $Z \circ X$. Then the composition of the corresponding Markov kernels satisfies:

$$\delta(Z \circ X | T) = \delta(Z | X) \circ \delta(X | T).$$

So the composition of Markov kernels extends the composition of (deterministic) functions.

(iii) *Composition of Gaussian Markov kernels.*

Proof. (ii) Let $\delta(Z \in C | X = x) := \mathbb{1}_C(Z(x))$ and $\delta(X \in A | T = t) := \mathbb{1}_A(X(t))$. Then

$$\begin{aligned} \delta(Z \in C | X) \circ \delta(X | T = t) &= (\delta(Z | X) \circ \delta(X | T))(C | t) \\ &= \int \delta(Z \in C | X = x) \delta(X \in dx | T = t) \\ &= \int \mathbb{1}_{Z^{-1}(C)}(x) \delta(X \in dx | T = t) \\ &= \delta(X \in Z^{-1}(C) | T = t) \end{aligned}$$

$$\begin{aligned}
&= \mathbb{1}_{X^{-1}(Z^{-1}(C))}(t) \\
&= \mathbb{1}_C(Z(X(t))) \\
&= \delta(Z(X) \in C \mid T = t) \\
&= \delta(Z \circ X \in C \mid T = t) = \delta(Z \circ X \mid T)(C \mid t).
\end{aligned}$$

□

DEFINITION 2.35 (Push-forward Markov Kernel).

Let $(\mathcal{W} \times \mathcal{T}, K(W \mid T))$ be a transition probability space and

$$X : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{X}$$

be a conditional random variable. Then we define the *push-forward Markov kernel* $K(X \mid T)$ of $K(W \mid T)$ with symbols:

$$K(X \mid T) := X_* K(W \mid T) = K(X(W, T) \mid T),$$

via:

$$K(X \mid T) : \mathcal{T} \rightarrow \mathcal{X}, \quad (A, t) \mapsto K(X \in A \mid T = t) := K(W \in X_t^{-1}(A) \mid T = t),$$

where, again:

$$X_t^{-1}(A) = \left(X^{-1}(A) \right)_t := \{w \in \mathcal{W} \mid X(w, t) \in A\}.$$

EXAMPLE 2.36. (i) *Push-forwards as compositions:*

$$K(X \mid T) = \delta(X \mid W, T) \circ K(W \mid T).$$

(ii) *Markov kernel including T .*

Proof. (i) We define

$$\delta(X \in A \mid W = w, T = t) := \mathbb{1}_A(X(w, t)) = \mathbb{1}_{X^{-1}(A)}(w, t).$$

□

DEFINITION 2.37 (Product of Markov Kernels).

Consider two Markov kernels:

$$Q(Z \mid Y, W, T) : \mathcal{Y} \times \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{Z}, \quad K(W, U \mid T, X) : \mathcal{T} \times \mathcal{X} \rightarrow \mathcal{W} \times \mathcal{U}.$$

Then we define the product Markov kernel:

$$Q(Z \mid Y, W, T) \otimes K(W, U \mid T, X) : \mathcal{Y} \times \mathcal{T} \times \mathcal{X} \rightarrow \mathcal{Z} \times \mathcal{W} \times \mathcal{U},$$

using measurable sets $E \subseteq \mathcal{Z} \times \mathcal{W} \times \mathcal{U}$ via:

$$\iint \mathbb{1}_E(z, w, u) Q(Z \in dz \mid Y = y, W = w, T = t) K((W, U) \in d(w, u) \mid T = t, X = x),$$

where the inner integration is over $z \in \mathcal{Z}$ and the outer integration over $(w, u) \in \mathcal{W} \times \mathcal{U}$.

EXAMPLE 2.38 (Product of Discrete Markov Kernels). Let $Q(Z \mid Y, W, T)$ and $K(W \mid T, X)$ be two Markov kernels on finite spaces. Let

$$P(Z, W \mid Y, T, X) := Q(Z \mid Y, W, T) \otimes K(W \mid T, X)$$

be the product of Markov kernels and p, q, k the corresponding mass functions. Then we have:

$$p(z_i, w_k \mid y_s, x_l, t_j) = q(z_i \mid y_s, w_k, t_j) \cdot k(w_k \mid t_j, x_l),$$

which is just the product of mass functions.

EXAMPLE 2.39 (Bayesian Statistics). Let $P(X \mid \Theta)$ be a statistical model with parameters θ , and $\Pi(\Theta \mid T)$ be a family of priors with hyper-parameters t . Then the joint distribution of X, Θ given hyper-parameters $T = t$ is:

$$P(X, \Theta \mid T) := P(X \mid \Theta) \otimes \Pi(\Theta \mid T), \quad p(x, \theta \mid t) = p(x \mid \theta) \cdot \pi(\theta \mid t).$$

THEOREM 2.40 (Fubini's Theorem).

Let (X, μ) and (Y, ν) be two (σ) -finite measure spaces and $f : X \times Y \rightarrow [0, \infty]$ a $(\mathcal{B}_X \otimes \mathcal{B}_Y)$ - $\mathcal{B}_{[0, \infty]}$ -measurable map. Then we have the equalities:

$$\int \left(\int f(x, y) \mu(dx) \right) \nu(dy) = \int \left(\int f(x, y) \nu(dy) \right) \mu(dx) = \int f(x, y) d(\mu \otimes \nu)(x, y).$$

Remark. For $D \in \mathcal{B}_X \otimes \mathcal{B}_Y$ Fubini's theorem says:

$$\int \left(\int \mathbb{1}_D(x, y) \mu(dx) \right) \nu(dy) = \int \left(\int \mathbb{1}_D(x, y) \nu(dy) \right) \mu(dx) = \int \mathbb{1}_D(x, y) d(\mu \otimes \nu)(x, y).$$

The integral notation hides the fact that $\mu \otimes \nu$ and $\nu \otimes \mu$ can only be identified as measures if we also swap the order of the spaces X and Y . $\mu \otimes \nu$ lives on $X \times Y$ and $\nu \otimes \mu$ lives on $Y \times X$. In more precise terms, we would have:

$$(\mu \otimes \nu)(D) = (\nu \otimes \mu)(D^s),$$

where $D^s := \{(y, x) \in Y \times X \mid (x, y) \in D\} \in \mathcal{B}_Y \otimes \mathcal{B}_X$. That said, we will always make this swap implicitly, and just write:

$$Q(Z \mid Y, T) \otimes K(U \mid T, X) = K(U \mid T, X) \otimes Q(Z \mid Y, T).$$

DEFINITION 2.41 (Marginalising Markov Kernels).

Let

$$K(X, Y \mid T) : \mathcal{T} \rightarrow X \times Y$$

be a Markov kernel in two variables. We can then define the *marginal Markov kernels* as follows:

$$\begin{aligned} K(X \mid T) : \mathcal{T} &\rightarrow X, \quad (A, t) \mapsto K(X \in A, Y \in Y \mid T = t), \\ K(Y \mid T) : \mathcal{T} &\rightarrow Y, \quad (B, t) \mapsto K(X \in X, Y \in B \mid T = t). \end{aligned}$$

EXAMPLE 2.42 (Marginal Markov Kernels of Discrete Markov Kernels). Let

$$K(X, Y \mid T) : \mathcal{T} \rightarrow X \times Y$$

be a Markov kernel in two variables on discrete spaces and $k_{X, Y \mid T}$ its mass function. We can then compute the marginal Markov kernels as follows:

$$k_{X \mid T}(x \mid t) = \sum_{y \in Y} k_{X, Y \mid T}(x, y \mid t),$$

$$k_{Y|T}(y | t) = \sum_{x \in \mathcal{X}} k_{X,Y|T}(x, y | t).$$

Note, by abuse of notation, for simplicity, we often omit the indices and write $k(x | t)$ and $k(y | t)$ instead and distinguish these two functions just by the use of the argument symbols x and y .

Remark.

DEFINITION 2.43 (Disintegration of Markov Kernels).

Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be measurable spaces where $\mathcal{X}$ and $\mathcal{Y}$ are standard measurable spaces. Let

$$K(X, Y | Z) : \mathcal{Z} \rightarrow \mathcal{X} \times \mathcal{Y}$$

be a Markov kernel and $K(Y | Z)$ its marginal Markov kernel given by

$$K(Y \in B | Z) = K(X \in \mathcal{X}, Y \in B | Z) : \mathcal{Z} \rightarrow \mathcal{Y}.$$

Then there exists a *essentially unique* Markov kernel (called *conditional Markov kernel*):

$$K(X | Y, Z) : \mathcal{Y} \times \mathcal{Z} \rightarrow \mathcal{X}$$

such that:

$$K(X, Y | Z) = K(X | Y, Z) \otimes K(Y | Z).$$

Remark. If $Q(X | Y, Z)$ is another Markov kernel then we have:

$$K(X, Y | Z) = Q(X | Y, Z) \otimes K(Y | Z),$$

if and only if the measurable subset N of $\mathcal{Y} \times \mathcal{Z}$ defined via

$$N := \{(y, z) \in \mathcal{Y} \times \mathcal{Z} \mid \exists A \in \mathcal{B}_X : Q(X \in A | Y = y, Z = z) \neq K(X \in A | Y = y, Z = z)\}$$

is a $K(Y | Z)$ -null set in $\mathcal{Y} \times \mathcal{Z}$.

EXAMPLE 2.44 (Conditional Markov Kernel for Discrete Markov Kernels). Consider a Markov kernel $K(X, Y | Z)$ where all spaces are discrete and let k be the corresponding mass function. Then the marginal mass functions are given by:

$$k(y | z) = \sum_{x \in \mathcal{X}} k(x, y | z), \quad k(x | z) = \sum_{y \in \mathcal{Y}} k(x, y | z).$$

A conditional Markov kernel conditioned on Y can then be defined via the mass function:

$$k(x | y, z) := \begin{cases} \frac{k(x, y | z)}{k(y | z)} & \text{if } k(y | z) > 0, \\ k(x | z) & \text{if } k(y | z) = 0. \end{cases}$$

With this setting we then have for all values x, y, z :

$$k(x, y | z) = k(x | y, z) \cdot k(y | z).$$

THEOREM 2.45 (Existence of Conditional Markov Kernels).

Let

$$K(X, Y | Z) : \mathcal{Z} \rightarrow \mathcal{X} \times \mathcal{Y}$$

be a Markov kernel with standard measurable spaces $\mathcal{X}, \mathcal{Y}$ (e.g. $\mathbb{R}^D, \mathbb{R}^K$) and $K(Y | Z)$ its marginal. Then there always exists a conditional Markov kernel $K(X | Y, Z)$ and it is essentially uniquely given by its defining equation:

$$K(X, Y | Z) = K(X | Y, Z) \otimes K(Y | Z).$$

Proof.

□

EXAMPLE 2.46 (Bayesian Statistics).

THEOREM 2.47.

Let $\mathcal{Z}$ be any measurable space and $\mathcal{X} = \bar{\mathbb{R}} = [-\infty, \infty]$. Let $K(X | Z) : \mathcal{Z} \dashrightarrow \mathcal{X}$ be a Markov kernel, $P(E)$ be the uniform distribution on $\mathcal{E} := [0, 1]$ and:

$$R(e | z) := \inf\{\tilde{x} \in \mathcal{X} \mid K(X \leq \tilde{x} | z) \geq e\},$$

the (conditional) quantile function (or inverse cumulative distribution function) of $K(X | Z)$. Then we can write $K(X | Z)$ as the push-forward:

$$K(X | Z) = \delta(R | E, Z) \circ P(E).$$

More explicitly, for $A \in \mathcal{B}_X$ and $z \in \mathcal{Z}$ we have:

$$K(X \in A | Z = z) = P(R(E | z) \in A).$$

SECTION 2.5

| Conditional Independence

If we throw two dice with outcome values X and Y , respectively, then knowing the value of Y does not give us any information about the value of X , and vice versa. We say that X and Y are independent from each other.

DEFINITION 2.48 (Independence).

Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space and $X : \mathcal{W} \rightarrow \mathcal{X}$ and $Y : \mathcal{W} \rightarrow \mathcal{Y}$ be two random variables. We say that X and Y are *independent* if the following equation holds:

$$P(X, Y) = P(X) \otimes P(Y)$$

where $P(X, Y)$ is the joint and $P(X)$ and $P(Y)$ are the corresponding marginal distributions. In symbols we would write this as:

$$X \perp_{P(\mathcal{W})} Y.$$

LEMMA 2.49.

Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space, $\mathcal{X}$ and $\mathcal{Y}$ standard measurable spaces and $X : \mathcal{W} \rightarrow \mathcal{X}$ and $Y : \mathcal{W} \rightarrow \mathcal{Y}$ be two random variables. Then the following statements are equivalent:

- (i) $X \perp Y$ under $P(\mathcal{W})$;
- (ii) $P(X, Y) = P(X) \otimes P(Y)$;
- (iii) There exists a probability distribution $Q(X)$ such that $P(X, Y) = Q(X) \otimes P(Y)$;
- (iv) $P(X | Y) = P(X)$ holds $P(Y)$ -a.s.;
- (v) For all $A \in \mathcal{B}_X$ we have $\mathbb{E}[\mathbb{1}_A(X) | Y] = \mathbb{E}[\mathbb{1}_A(X)]$, $P(\mathcal{W})$ -a.s.;
- (vi) For all $A \in \mathcal{B}_X$ and $B \in \mathcal{B}_Y$ we have $P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B)$.

EXAMPLE 2.50. Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space, X, Y two discrete random variables. Then $X \perp Y$ is equivalent to $\forall x \in \mathcal{X}, \forall y \in \mathcal{Y}, p(x, y) = p(x) \cdot p(y)$.

DEFINITION 2.51 (Mutual Independence).

Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space and I an (index) set. For $i \in I$ let $X_i : \mathcal{W} \rightarrow \mathcal{X}_i$ be a random variable. We say that $(X_i)_{i \in I}$ is (*mutually/jointly*) *independent* if for all two disjoint subsets $J_1 \cup J_2 \subseteq I$ we have the independence:

$$(X_{j_1})_{j_1 \in J_1} \perp_{P(\mathcal{W})} (X_{j_2})_{j_2 \in J_2}.$$

EXAMPLE 2.52. Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space. A finite tuple of random variables $(X_1, \dots, X_n)$ is mutually independent if and only if

$$P(X_1, \dots, X_n) = P(X_1) \otimes \dots \otimes P(X_n).$$

EXAMPLE 2.53 (Coin Flips). (i) Consider two independent coin flips with outcome variables X and Y , respectively, with values in $\{0, 1\}$, and $Z := X + Y \in \{0, 1, 2\}$. If the value of Z is known, say $Z = 1$, then revealing the value of Y , say $Y = 0$, provides us with all the information to fully determine the value of X , here $X = 1$. This is despite the fact

that X and Y were assumed to be independent. This means that conditioning on a third variable Z can destroy independence. In this case, we say that X and Y are dependent conditioned on Z . Summarised in symbols we have:

$$X \perp_{P(W)} Y, \quad X \not\perp_{P(W)} Y \mid Z.$$

- (ii) Now consider three (mutually) independent coin flips X, W, U with values in $\{0, 1\}$. Let $Z := X + W$ and $Y := Z + U = X + W + U$. If we know the value of Y , say $Y = 0$, then we would have information about the values of X as well, here $X = 0$. This shows that X and Y cannot be independent random variables. If, in contrast, we would first reveal the value of Z , say $Z = 1$, then the value of X might be restricted by the value of Z , but also revealing Y would not give us any *additional* information about the value of X . The reason is that $Y = Z + U$ and U is independent of X, W and $Z = X + W$. So, even though X and Y are dependent, when conditioned on Z they become independent, as there is no additional information gained about each other's value when revealing the other. Summarised in symbols we have:

$$X \not\perp_{P(W)} Y, \quad X \perp_{P(W)} Y \mid Z.$$

DEFINITION 2.54 (Conditional Independence for Random Variables).

Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space and $\mathcal{X}, \mathcal{Y}$ and $\mathcal{Z}$ standard measurable spaces, and $X : \mathcal{W} \rightarrow \mathcal{X}$ and $Y : \mathcal{W} \rightarrow \mathcal{Y}$ and $Z : \mathcal{W} \rightarrow \mathcal{Z}$ be three random variables. We then say that X is independent of Y conditioned on Z if any of the following equivalent conditions holds:

- (i) $P(X, Y \mid Z) = P(X \mid Z) \otimes P(Y \mid Z)$ holds $P(Z)$ -a.s.;
- (ii) $P(X \mid Y, Z) = P(X \mid Z)$ holds $P(Y, Z)$ -a.s.;
- (iii) there exists a Markov kernel $Q(X \mid Z) : \mathcal{Z} \rightarrow \mathcal{X}$ such that $P(X, Y, Z) = Q(X \mid Z) \otimes P(Y, Z)$.
- (iv) for every $A \in \mathcal{B}_X$ and $B \in \mathcal{B}_Y$ we have $\mathbb{E}[\mathbb{1}_A(X) \cdot \mathbb{1}_B(Y) \mid Z] = \mathbb{E}[\mathbb{1}_A(X) \mid Z] \cdot \mathbb{E}[\mathbb{1}_B(Y) \mid Z]$, $P(W)$ -a.s.;
- (v) for every $A \in \mathcal{B}_X$ we have $\mathbb{E}[\mathbb{1}_A(X) \mid Y, Z] = \mathbb{E}[\mathbb{1}_A(X) \mid Z]$, $P(W)$ -a.s.

Remark. In those cases, in symbols we write

$$X \perp_{P(W)} Y \mid Z.$$

EXAMPLE 2.55. Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space and X, Y, Z be discrete random variables. Then $X \perp Y \mid Z$ is equivalent to

$$p(x, y \mid z) = p(x \mid z) \cdot p(y \mid z)$$

for all $x \in \mathcal{X}$, $y \in \mathcal{Y}$, and $z \in \mathcal{Z}$ such that $p(z) > 0$.

THEOREM 2.56 (Separoid Axioms).

Let $(\mathcal{W}, P(\mathcal{W}))$ be a probability space and X, Y, Z and U random variables taking values in standard measurable spaces $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ and $\mathcal{U}$, respectively. Then we have the following rules.

- (i) *Redundancy:* if $U = \varphi(X)$ a.s. is a measurable function of X , e.g. $U = X$, then $U \perp_{P(W)} Y \mid X$.

- (ii) *Symmetry*: $X \perp_{P(W)} Y \mid Z \implies Y \perp_{P(W)} X \mid Z$.
- (iii) *Decomposition*: $X \perp_{P(W)} Y, U \mid Z \implies X \perp_{P(W)} U \mid Z$.
- (iv) *Weak Union*: $X \perp_{P(W)} Y, U \mid Z \implies X \perp_{P(W)} Y \mid U, Z$.
- (v) *Contraction*: $\left(X \perp_{P(W)} U \mid Z \right) \wedge \left(X \perp_{P(W)} Y \mid U, Z \right) \implies X \perp_{P(W)} Y, U \mid Z$.

Proof. □

EXAMPLE 2.57. Consider a probabilistic program with input variables T, S and output variables X, Y, Z . Whenever the program is given T and S as input, it internally samples $U, E \sim U[0, 1]$ uniformly and independently from a random number generator, then calculates

$$X := T + S + U, \quad Y := 5 \cdot S + E, \quad Z := X \cdot Y,$$

and, finally, outputs X, Y and Z . Even though the input T and S is provided by the user and is not considered a random variable, we can reason about the fact that “Output Y only depends on the input S and not on the input T .” We want to formalise such conditional independence mathematically in order to be able to write this as:

$$Y \perp T \mid S.$$

DEFINITION 2.58 (Conditional Independence for Conditional Random Variables).

Let $(\mathcal{W} \times \mathcal{T}, K(W \mid T))$ be a transition probability space with Markov kernel:

$$K(W \mid T) : \mathcal{T} \rightarrow \mathcal{W}.$$

Consider conditional random variables:

$$X : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{X}, \quad Y : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{Y}, \quad Z : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{Z}.$$

We say that X is independent of Y conditioned on Z w.r.t. $K(W \mid T)$, in symbols:

$$X \perp_{K(W|T)} Y \mid Z,$$

if there exists a Markov kernel:

$$Q(X \mid Z) : \mathcal{Z} \rightarrow \mathcal{X},$$

such that:

$$K(X, Y, Z \mid T) = Q(X \mid Z) \otimes K(Y, Z \mid T),$$

where $K(Y, Z \mid T)$ is the marginal of $K(X, Y, Z \mid T)$.

Remark. (i) The Markov kernel $Q(X \mid Z)$ appearing in the conditional independence

$$X \perp_{K(W|T)} Y \mid Z$$

is then a version of a conditional Markov kernel $K(X \mid Y, Z, T)$ and is thus essentially unique. We will use the following suggestive notation for it:

$$K(X \mid \mathcal{T}, \mathcal{Y}, Z) := Q(X \mid Z),$$

and does not depend on arguments y and t .

(ii) We have the equivalence:

$$X \perp_{K(W|T)} Y | Z \iff X \perp_{K(W|T)} T, Y | Z,$$

where $T : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{T}$, $(w, t) \mapsto t$, is the canonical projection map.

NOTATION 2.59 (Conditional Independence for Random Variables).

We recover the notion of conditional independence for random variables X, Y, Z with standard measurable spaces as codomains by taking $\mathcal{T} := \{*\}$, $P(W) := K(W | *)$:

$$X \perp_{P(W)} Y | Z \iff \exists Q(X | Z) : P(X, Y, Z) = Q(X | Z) \otimes P(Y, Z).$$

Such a $Q(X | Z)$ is then a conditional probability distribution of $P(X, Z)$ conditioned on Z . In suggestive notations:

$$Q(X | Z) =: P(X | Y, Z) = P(X | Z).$$

EXAMPLE 2.60 (Conditional Independence for Deterministic Mappings). If for example, $\mathcal{T} = \mathcal{T}_1 \times \mathcal{T}_2$ and $T_i : \mathcal{W} \times \mathcal{T}_1 \times \mathcal{T}_2 \rightarrow \mathcal{T}_i$ the canonical projection onto $\mathcal{T}_i$, then F is a function in two variables (t_1, t_2) . We then have:

$$F \perp_{K(W|T)} T_1 | T_2$$

if and only if F —as a function—is only dependent on the argument t_2 (and not on t_1).

EXAMPLE 2.61 (Statistical Concepts). Let $P(W | \Theta)$ be a statistical model, considered as a Markov kernel $\mathcal{F} \rightarrow \mathcal{W}$. Let X and Y be two conditional random variables *w.r.t.* $P(W | \Theta)$. A statistic of X is a measurable map $S : \mathcal{X} \rightarrow \mathcal{S}$, which we consider as the conditional random variable $S \preceq X$ given via:

- (i) S is an *ancillary statistic* of X *w.r.t.* Θ if and only if $S \perp_{P(W|\Theta)} \Theta$;
- (ii) S is a *sufficient statistic* of X *w.r.t.* Θ if and only if $X \perp_{P(W|\Theta)} \Theta | S$;
- (iii) S is an *adequate statistic* of X for Y *w.r.t.* Θ if and only if $X \perp_{P(W|\Theta)} \Theta, Y | S$.

SECTION 2.6

| Separoid Axioms for Conditional Independence

THEOREM 2.62 ((Asymmetric) Separoid Axioms for Conditional Independence).

Let $(\mathcal{W} \times \mathcal{T}, K(W | T))$ be a transition probability space with Markov kernel:

$$K(W | T) : \mathcal{T} \rightarrow \mathcal{W}.$$

Consider conditional random variables X, Y, Z, U with common domain $\mathcal{W} \times \mathcal{T}$ and standard measurable spaces⁴ $\mathcal{X}, \mathcal{Y}, \mathcal{Z}, \mathcal{U}$, respectively, as codomains. Let

$$T : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{T}$$

be the canonical projection and $*$ the constant conditional random variable. We write $U \preceq X$ if there exists a measurable function $G : \mathcal{X} \rightarrow \mathcal{U}$ such that $U = G \circ X$. Then the ternary relation $\perp = \perp_{K(W|T)}$ satisfies the following rules.

(i) Extended Left Redundancy:

$$U \preceq X \implies U \perp Y | X.$$

(ii) T-Restricted Right Redundancy:

$$X \perp * | Z, T$$

always holds.

(iii) T-Inverted Right Decomposition:

$$X \perp Y | Z \implies X \perp T, Y | Z.$$

(iv) Left Decomposition:

$$X, U \perp Y | Z \implies U \perp Y | Z.$$

(v) Right Decomposition:

$$X \perp Y, U | Z \implies X \perp U | Z.$$

(vi) Left Weak Union:

$$X, U \perp Y | Z \implies X \perp Y | U, Z.$$

(vii) Right Weak Union:

$$X \perp Y, U | Z \implies X \perp Y | U, Z.$$

(viii) Left Contraction:

$$(X \perp Y | U, Z) \wedge (U \perp Y | Z) \implies X, U \perp Y | Z.$$

(ix) Right Contraction:

$$(X \perp Y | U, Z) \wedge (X \perp U | Z) \implies X \perp Y, U | Z.$$

(x) Right Cross Contraction:

$$(X \perp Y | U, Z) \wedge (U \perp X | Z) \implies X \perp Y, U | Z.$$

⁴ T-Restricted Right Redundancy, Left Weak Union and Symmetry need the existence of conditional Markov kernels. That is the reason we assumed standard measurable spaces.

(xi) *Flipped Left Cross Contraction:*

$$(X \perp Y \mid U, Z) \wedge (Y \perp U \mid Z) \implies Y \perp X, U \mid Z.$$

In particular, we have the equivalences:

$$(X \perp Y, U \mid Z) \iff (X \perp Y \mid U, Z) \wedge (X \perp U \mid Z),$$

$$(X, U \perp Y \mid Z) \iff (X \perp Y \mid U, Z) \wedge (U \perp Y \mid Z).$$

(xii) *T-Restricted Symmetry:*

$$X \perp Y \mid Z, T \implies Y \perp X \mid Z, T.$$

In the special case of $\mathcal{T} = \{\}$, the one-point space, (i.e., in the case of probability distributions and random variables mapping to standard measurable spaces) we thus have (unrestricted) Symmetry.*

Exercises

EXERCISE 2.1 (Push-Forward Distributions). (a) Let X be uniform on $\{-M, -M+1, \dots, N\}$, where $M, N \in \mathbb{N}$ and $M > N$. Find the probability mass function of $Y = |X|$.

(b) Let $X \sim \mathcal{N}(0, 1)$. Find the Lebesgue density of $Y = \exp(X)$.

EXERCISE 2.2 (Expected Values of Dice). A discrete die is uniform on $\mathcal{X} = \{a, b, c, d, e, f\}$, with value map $V_D(a) = 1, \dots, V_D(f) = 6$. A continuous die is uniform on $[0, 1]$, with value map $V_C(x) = 5x + 1$. Compute both expected values using measure-theoretic notation.

EXERCISE 2.3 (Measurability and Induced Sigma-Algebras). (a) For measurable spaces $(A, \mathcal{F}), (B, \mathcal{G}), (C, \mathcal{H})$, prove that the composition of an $\mathcal{F}$ - $\mathcal{G}$ measurable $f : A \rightarrow B$ and a $\mathcal{G}$ - $\mathcal{H}$ measurable $g : B \rightarrow C$ is $\mathcal{F}$ - $\mathcal{H}$ measurable.

(b) For an arbitrary set U , measurable space (V, Σ) and map $X : U \rightarrow V$, prove that $\Sigma_X = \{X^{-1}(A) : A \in \Sigma\}$ is a sigma-algebra and is the smallest sigma-algebra making X measurable. In particular, show $\Sigma_X \subseteq \mathcal{F}$ whenever X is $\mathcal{F}$ - Σ measurable.

(c) Show that $\mathbb{1}_A : \Omega \rightarrow \{0, 1\}$ is $\mathcal{B}_\Omega$ - $2^{\{0,1\}}$ measurable whenever $A \in \mathcal{B}_\Omega$.

(d) For Lebesgue measure λ on $(\mathbb{R}, \mathcal{B}_\mathbb{R})$, prove $\lambda(\{1, 1/2, 1/3, \dots\}) = 0$.

EXERCISE 2.4 (Scaling Expected Values). (a) For a simple random variable $X : \Omega \rightarrow \mathbb{R}$, prove $\mathbb{E}[cX] = c\mathbb{E}[X]$ directly, without invoking linearity of the integral.

(b) Extend the argument to a general real random variable whenever the expectations are defined.

EXERCISE 2.5 (Testing Measurability on Generators). Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a function, $\mathcal{B}_\mathcal{X}$ a sigma-algebra, and $\mathcal{B}$ a collection of subsets of $\mathcal{Y}$. Assume $f^{-1}(B) \in \mathcal{B}_\mathcal{X}$ for every $B \in \mathcal{B}$. Prove that f is measurable into $\mathcal{B}_\mathcal{Y} = \sigma(\mathcal{B})$. *Hint. Show $\mathcal{B} \subseteq f_*\mathcal{B}_\mathcal{X}$, where*

$$f_*\mathcal{B}_\mathcal{X} = \{B \subseteq \mathcal{Y} : f^{-1}(B) \in \mathcal{B}_\mathcal{X}\},$$

then deduce $\sigma(\mathcal{B}) \subseteq f_\mathcal{B}_\mathcal{X}$.*

SOLUTION. By assumption, $f^{-1}(B) \in \mathcal{B}_\mathcal{X}$ for every $B \in \mathcal{B}$, so $\mathcal{B} \subseteq f_*\mathcal{B}_\mathcal{X}$. Pre-images preserve complements and countable unions, hence $f_*\mathcal{B}_\mathcal{X}$ is a σ -algebra. Therefore $\mathcal{B}_\mathcal{Y} = \sigma(\mathcal{B}) \subseteq f_*\mathcal{B}_\mathcal{X}$. For every $C \in \mathcal{B}_\mathcal{Y}$, this gives $f^{-1}(C) \in \mathcal{B}_\mathcal{X}$, proving measurability. $\square$

EXERCISE 2.6 (Evaluation Maps). For measurable spaces $(\mathcal{X}, \mathcal{B}_\mathcal{X})$ and $(\mathcal{Y}, \mathcal{B}_\mathcal{Y})$, prove that $K : \mathcal{X} \rightarrow \mathcal{M}(\mathcal{Y})$ is measurable into the evaluation sigma-algebra if and only if $ev_B \circ K : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ is measurable for every $B \in \mathcal{B}_\mathcal{Y}$.

EXERCISE 2.7 (Interpreting Markov Kernel Operations). Two players play three chess games. Let $X, Y, Z \in \{1, 2\}$ denote the winner of each game and $C \in \{1, 2\}$ the player selected by a coin toss to play white first. After each game, the losing player plays white next, with an advantage over black. Interpret:

- (a) $N(Z | Y), M(Y | X), Q(X | C)$ and $K(C | *)$;
- (b) $N(Z | Y) \circ (M(Y | X) \circ Q(X | C))$;
- (c) $P(Z, Y, X | C) = N(Z | Y) \otimes (M(Y | X) \otimes Q(X | C))$;
- (d) $P(Y | C)$, also expressing it as a composition;
- (e) $P(Z, X | Y, C)$ and its relation to the product and marginal above;
- (f) $f_*P(Z, Y, X | C)$, where Player 1's prize is

$$f(z, y, x, c) = \begin{cases} 100 - 10\mathbb{1}_{\{1\}}(c), & z + y + x \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

EXERCISE 2.8 (Associativity and Commutativity). Consider kernels $K(W, U | T, X), Q(Z | Y, W, T)$ and $P(H | Z, T)$ on the corresponding product spaces.

- (a) Prove

$$\begin{aligned} P(H | Z, T) \otimes [Q(Z | Y, W, T) \otimes K(W, U | T, X)] \\ = [P(H | Z, T) \otimes Q(Z | Y, W, T)] \otimes K(W, U | T, X). \end{aligned}$$

- (b) Explain why $Q(Z | Y, W, T) \otimes K(W, U | T, X)$ generally does not commute, and state conditions under which products commute.
- (c) State conditions under which compositions of Markov kernels commute.

EXERCISE 2.9 (Operations on Deterministic Kernels). Let $\mathcal{T}$ be measurable and $\mathcal{X}, \mathcal{Y}$ standard measurable. For measurable $f_1 : \mathcal{W} \rightarrow \mathcal{T}, f_2 : \mathcal{T} \rightarrow \mathcal{X}$ and $f_3 : \mathcal{T} \rightarrow \mathcal{Y}$, specify the domain and codomain of each kernel, express its deterministic function in terms of f_1, f_2, f_3 , and justify the result:

- (a) $\delta_{f_c} = \delta_{f_2} \circ \delta_{f_1}$;
- (b) $\delta_{f_p} = \delta_{f_2} \otimes \delta_{f_3}$;
- (c) the $\mathcal{X}$ -marginal δ_{f_m} of δ_{f_p} ;
- (d) a conditional $\delta_{f_{cd}}$ satisfying $\delta_{f_p} = \delta_{f_{cd}} \otimes \delta_{f_m}$.

EXERCISE 2.10 (Linear Gaussian Kernel Parameters). Let $k(w | t) = \mathcal{N}(w | At + b, \Sigma)$ and $q(z | w) = \mathcal{N}(z | Cw + d, \Gamma)$ be densities of kernels $K(W | T)$ and $Q(Z | W)$.

- (a) Find the parameters of $Q(Z | W) \otimes K(W | T)$.
- (b) Find the parameters of $Q(Z | W) \circ K(W | T)$.
- (c) Find the parameters of the conditional $P(W | Z, T)$ of the product.

Suggested derivation. Recall the density formula for composition; express $\log p(z, w | t)$ as a quadratic polynomial with block precision matrix; invert it to find the covariance; complete the square; and read off the marginal. For symmetric M, N , the identity

$$\begin{pmatrix} N & -NT \\ -T^TN & M + T^TN T \end{pmatrix}^{-1} = \begin{pmatrix} N^{-1} + TM^{-1}T^T & TM^{-1} \\ M^{-1}T^T & M^{-1} \end{pmatrix}$$

may be used without proof. State the external source used for Gaussian conditioning and explain how its notation translates to this problem.

SOLUTION. Let $K(W | T)$ and $Q(Z | W)$ have densities

$$k(w | t) = \mathcal{N}(w | At + b, \Sigma), \quad q(z | w) = \mathcal{N}(z | Cw + d, \Gamma),$$

with positive definite Σ and Γ . Write $m_t = At + b$ and $S = \Gamma + C\Sigma C^T$. Their product is Gaussian:

$$p(z, w | t) = \mathcal{N}\left(\begin{pmatrix} z \\ w \end{pmatrix} \middle| \begin{pmatrix} Cm_t + d \\ m_t \end{pmatrix}, \begin{pmatrix} S & C\Sigma \\ \Sigma C^T & \Sigma \end{pmatrix}\right).$$

The composition is its Z -marginal:

$$p(z | t) = \int q(z | w)k(w | t)dw = \mathcal{N}(z | Cm_t + d, S).$$

The conditional density is

$$p(w | z, t) = \mathcal{N}\left(w | m_t + \Sigma C^T S^{-1}(z - Cm_t - d), \Sigma - \Sigma C^T S^{-1} C \Sigma\right).$$

The log product density is

$$\begin{aligned} \log p(z, w | t) &= -\frac{1}{2}(w - m_t)^T \Sigma^{-1}(w - m_t) \\ &\quad - \frac{1}{2}(z - Cw - d)^T \Gamma^{-1}(z - Cw - d) + \text{constant}. \end{aligned}$$

Its precision matrix and linear term are

$$\Lambda = \begin{pmatrix} \Gamma^{-1} & -\Gamma^{-1}C \\ -C^T\Gamma^{-1} & \Sigma^{-1} + C^T\Gamma^{-1}C \end{pmatrix}, \quad h = \begin{pmatrix} \Gamma^{-1}d \\ \Sigma^{-1}m_t - C^T\Gamma^{-1}d \end{pmatrix}.$$

Block inversion gives the displayed covariance, and completing the square gives the mean $\Lambda^{-1}h = (Cm_t + d, m_t)^T$. Completing the square in w also yields

$$\begin{aligned} H &= \Sigma^{-1} + C^T\Gamma^{-1}C, \\ \text{Cov}(W | z, t) &= H^{-1}, \\ \mathbb{E}[W | z, t] &= H^{-1}(\Sigma^{-1}m_t + C^T\Gamma^{-1}(z - d)). \end{aligned}$$

These equal the conditional covariance and mean above. $\square$

EXERCISE 2.11 (The Product Is a Markov Kernel). For $Q(Z | W, T) : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{Z}$ and $K(W | T) : \mathcal{T} \rightarrow \mathcal{W}$, show that their product induces

$$M : \mathcal{B}_{\mathcal{Z} \times \mathcal{W}} \times \mathcal{T} \rightarrow [0, 1].$$

Prove that $D \mapsto M(D, t)$ is a probability measure for every t and $t \mapsto M(D, t)$ is measurable for every measurable D .

EXERCISE 2.12 (Conditional Independence from a Table). For the following unnormalised joint mass function, determine whether $X \perp Y | Z$. Give the derivation.

z	x	$y = 1$	$y = 2$
1	1	2	4
1	2	3	6
1	3	1	2
2	1	12	12
2	2	6	6
2	3	0	0

EXERCISE 2.13 (Discrete Separoid Axioms). For discrete X, Y, Z, U with joint mass function p , prove:

- (a) redundancy: $X \perp Y | X$;
- (b) symmetry: $X \perp Y | Z \Rightarrow Y \perp X | Z$;
- (c) left decomposition: $X, U \perp Y | Z \Rightarrow U \perp Y | Z$;
- (d) left weak union: $X, U \perp Y | Z \Rightarrow X \perp Y | U, Z$;
- (e) left contraction: $(X \perp Y | U, Z) \wedge (U \perp Y | Z) \Rightarrow X, U \perp Y | Z$;
- (f) under strictly positive p , intersection:

$$(X \perp Y | U, Z) \wedge (X \perp U | Y, Z) \Rightarrow X \perp U, Y | Z.$$

EXERCISE 2.14 (Generalised Inverses and Kernel Sampling). Let $F(x) = P(X \leq x)$ and $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. Draw a discontinuous distribution function and its generalised inverse. Recall $F^{-1}(F(X)) = X$ almost surely; $F(X)$ is uniform when F is continuous; and $F^{-1}(U) \sim P(X)$ for uniform U .

- (a) For the Gaussian kernel with density

$$\mathcal{N}(x | t_1, t_2^2) = \frac{1}{\sqrt{2\pi}t_2} \exp\left(-\frac{(x - t_1)^2}{2t_2^2}\right), \quad t_2 > 0,$$

find a deterministic map whose push-forward of a constant uniform kernel on $[0, 1]$ is this kernel. Inverse error functions may be used.

- (b) Define the kernel with rate $\lambda > 0$ and density $\mathbb{1}_{[0, \infty)}(x)\lambda e^{-\lambda x}$. Find its deterministic uniform sampling map.

SOLUTION. For a distribution function $F(x) = P(X \leq x)$, its generalised inverse is

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}, \quad 0 < u < 1.$$

Then $F^{-1}(F(X)) = X$ almost surely; if F is continuous, $F(X) \sim \mathcal{U}[0, 1]$; and $F^{-1}(U) \sim P(X)$ for $U \sim \mathcal{U}[0, 1]$. By [theorem 2.47](#), the same construction applies to Markov kernels.

For $t_1 \in \mathbb{R}$ and $t_2 > 0$, the Gaussian kernel

$$K(X \in D \mid t_1, t_2) = \int_D \frac{1}{\sqrt{2\pi}t_2} \exp\left(-\frac{(x - t_1)^2}{2t_2^2}\right) dx$$

is the push-forward of $U \sim \mathcal{U}[0, 1]$ under

$$R(u \mid t_1, t_2) = t_1 + t_2 \Phi^{-1}(u) = t_1 + \sqrt{2}t_2 \operatorname{erf}^{-1}(2u - 1).$$

For $\lambda > 0$, the exponential kernel is

$$K(X \in D \mid \lambda) = \int_D \mathbb{1}_{[0, \infty)}(x) \lambda e^{-\lambda x} dx,$$

with sampling map

$$R(u \mid \lambda) = -\frac{\log(1 - u)}{\lambda}.$$

The maps may be assigned arbitrary finite values at endpoints where the quantiles are infinite, since these endpoints have uniform probability zero.

The Gaussian distribution function is

$$F(x \mid t_1, t_2) = \Phi\left(\frac{x - t_1}{t_2}\right) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x - t_1}{\sqrt{2}t_2}\right) \right].$$

Solving $F(x \mid t_1, t_2) = u$ gives the Gaussian sampling map. For the exponential kernel, $F(x \mid \lambda) = 1 - e^{-\lambda x}$ for $x \geq 0$; solving $F(x \mid \lambda) = u$ gives the exponential sampling map. In both cases,

$$K(X \in D \mid t) = P(R(U \mid t) \in D).$$

□

Graph Theory

DEFINITION 3.1 (Conditional Directed Mixed Graphs (CDMG)).

A conditional directed mixed graph (CDMG) $G = (J, V, E, L)$ consists of:

- (i) set of *input/action/intervention/context nodes* J ;
- (ii) set of *output nodes* V ;
- (iii) set of *directed edges* $E \subseteq (J \cup V) \times V$;
- (iv) set of *bi-directed edges* $L \subseteq V \times V$ such that $(v_1, v_2) \in L \Rightarrow (v_2, v_1) \in L$.

NOTATION 3.2.

Let $G = (J, V, E, L)$ be a CDMG. We will write:

- (i) $v \in G$ to mean $v \in J \cup V$;
- (ii) $v_1 \rightarrow v_2 \in G$ to mean $(v_1, v_2) \in E$;
- (iii) $v_2 \rightarrow v_1 \in G$ to mean $(v_2, v_1) \in E$;
- (iv) $v_1 \leftrightarrow v_2 \in G$ to mean $(v_1, v_2) \in L$.

DEFINITION 3.3 (Walks).

Let $G = (J, V, E, L)$ be a CDMG and $v, w \in G$.

- (i) A *general walk* from v_0 to v_3 could look like:

$$v_0 \rightarrow v_1 \leftarrow v_2 \leftrightarrow v_3.$$

The same node may appear multiple times in a walk, and a walk is called *path* if no node occurs more than once.

- (ii) A *directed walk* from v to w in G is of the form:

$$v = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_{n-1} \rightarrow v_n = w,$$

for some $n \geq 0$, where all arrowheads point in the direction of w and there are no arrowheads pointing back.

- (iii) A *bi-directed walk* from v to w in G is of the form:

$$v = v_0 \leftrightarrow v_1 \leftrightarrow \cdots \leftrightarrow v_{n-1} \leftrightarrow v_n = w,$$

for some $n \geq 0$, where all edges are bi-directed.

- (iv) A *collider walk* from v to w in G is of the form:

$$v = v_0 \leftrightarrow v_1 \leftrightarrow \cdots \leftrightarrow v_{n-1} \leftarrow v_n = w,$$

for some $n \geq 0$, where all nodes in between v and w have two arrowheads pointing towards them (also known as *collider*).

(v) A *fork/bifurcation* from v to w in G is of the form:

$$v = v_0 \leftarrow v_1 \leftarrow \cdots \leftarrow v_{k-1} \leftarrow^* v_k \rightarrow \cdots \rightarrow v_{n-1} \rightarrow v_n = w,$$

such that $v \neq w$, the walk contains both endnodes exactly once, every node has at most one arrowhead pointing towards it, and both endnodes have exactly one arrowhead pointing towards them.

DEFINITION 3.4 (Family Relationships).

Let $G = (J, V, E, L)$ be a CDMG, $v, w \in V$ and $A \subseteq J \cup V$ a subset of nodes. We then define:

(i) the set of *parents* of v in G :

$$\text{Pa}^G(v) := \{w \in G \mid w \rightarrow v \in G\};$$

the set of *parents* of A in G :

$$\text{Pa}^G(A) := \bigcup_{v \in A} \text{Pa}^G(v);$$

(ii) the set of *children* of v in G :

$$\text{Ch}^G(v) := \{w \in G \mid v \rightarrow w \in G\};$$

the set of *children* of A in G :

$$\text{Ch}^G(A) := \bigcup_{v \in A} \text{Ch}^G(v);$$

(iii) the set of *siblings* of v in G :

$$\text{Sib}^G(v) := \{w \in G \mid v \leftrightarrow w \in G\};$$

(iv) the set of *ancestors* of v in G :

$$\text{Anc}^G(v) := \{w \in G \mid \exists \text{ directed walk: } w \rightarrow \cdots \rightarrow v \in G\}.$$

Note: $v \in \text{Anc}^G(v)$. The set of *ancestors* of A in G :

$$\text{Anc}^G(A) := \bigcup_{v \in A} \text{Anc}^G(v).$$

Note: $A \subseteq \text{Anc}^G(A)$;

(v) the set of *descendants* of v in G :

$$\text{Desc}^G(v) := \{w \in G \mid \exists \text{ directed walk: } v \rightarrow \cdots \rightarrow w \in G\}.$$

Note: $v \in \text{Desc}^G(v)$. The set of *descendants* of A in G :

$$\text{Desc}^G(A) := \bigcup_{v \in A} \text{Desc}^G(v).$$

Note: $A \subseteq \text{Desc}^G(A)$;

(vi) the set of *non-descendants* of A in G :

$$\text{NonDesc}^G(A) := (J \cup V) \setminus \text{Desc}^G(A);$$

(vii) the *strongly connected component* of v in G :

$$\text{Sc}^G(v) := \text{Anc}^G(v) \cap \text{Desc}^G(v).$$

Note: $v \in \text{Sc}^G(v)$. The (union of) *strongly connected components* of A in G :

$$\text{Sc}^G(A) := \bigcup_{v \in A} \text{Sc}^G(v).$$

Note: $A \subseteq \text{Sc}^G(A)$;

(viii) the *district* of v in G :

$$\text{Dist}^G(v) := \{w \in G \mid \exists \text{ bi-directed walk: } v \leftrightarrow v_1 \leftrightarrow \dots \leftrightarrow v_{n-1} \leftrightarrow w \in G\}.$$

Note: $v \in \text{Dist}^G(v)$. The *district* of A in G :

$$\text{Dist}^G(A) := \bigcup_{v \in A} \text{Dist}^G(v).$$

Note: $A \subseteq \text{Dist}^G(A)$.

DEFINITION 3.5 (Acyclicity).

A CDMG $G = (J, V, E, L)$ is called *acyclic* if there does not exist any non-trivial directed walk from v to itself in G for any node $v \in G$.

EXAMPLE 3.6. A Conditional Directed Mixed Graph (CDMG) $G = (J, V, E, L)$ is called:

- (i) *Conditional Acyclic Directed Mixed Graph (CADMG)* if G is acyclic;
- (ii) *Directed Mixed Graph (DMG)* if $J = \emptyset$;
- (iii) *Acyclic Directed Mixed Graph (ADMG)* if G is acyclic and $J = \emptyset$;
- (iv) *Conditional Directed Graph (CDG)* if $L = \emptyset$;
- (v) *Directed Graph (DG)* if $J = \emptyset$ and $L = \emptyset$;
- (vi) *Conditional Directed Acyclic Graph (CDAG)* if G is acyclic and $L = \emptyset$;
- (vii) *Directed Acyclic Graph (DAG)* if G is acyclic, $J = \emptyset$ and $L = \emptyset$.

DEFINITION 3.7 (Topological Order).

Let $G = (J, V, E, L)$ be a CDMG. A *topological order* of G is a total order $<$ of $J \cup V$ such that for all $v, w \in G$:

$$v \in \text{Pa}^G(w) \implies v < w.$$

Equivalently, it can be described as an indexing of the nodes $J \cup V = \{v_1, \dots, v_K\}$ where parents always precede their children.

LEMMA 3.8.

A CDMG $G = (J, V, E, L)$ is *acyclic* if and only if it has a *topological order*.

Proof.

□

DEFINITION 3.9 (Predecessors).

Let $G = (J, V, E, L)$ be a CDMG and $<$ a total order of $J \cup V$. The set of predecessors of v in G are:

$$\text{Pred}_<^G(v) := \{w \in G \mid w < v\},$$

$$\text{Pred}_{\leq}^G(v) := \{w \in G \mid w \leq v\} \cup \{v\}.$$

SECTION 3.1

| σ -Separation

DEFINITION 3.10 (Colliders and Non-Colliders).

Let $G = (J, V, E, L)$ be a CDMG and π a walk in G :

$$\pi = (v_0 \mathrel{\ast\ast} \dots \mathrel{\ast\ast} v_n).$$

A node v_k , or more precisely, the position $k \in \{0, \dots, n\}$, on the walk π is called:

- (i) a *non-collider* on π , if there is at most one arrowhead pointing towards v_k , i.e., if it falls into one of the following cases:
 - *end-node*: $k \in \{0, n\}$;
 - *left chain*: $v_{k-1} \leftarrow v_k \leftarrow \ast v_{k+1}$;
 - *right chain*: $v_{k-1} \ast \rightarrow v_k \rightarrow v_{k+1}$;
 - *fork*: $v_{k-1} \leftarrow v_k \rightarrow v_{k+1}$;
- (ii) a *collider* on π , if it is of the form:

$$v_{k-1} \ast \rightarrow v_k \leftarrow \ast v_{k+1},$$

i.e., if there are two arrowheads pointing towards v_k on the walk π .

DEFINITION 3.11 (Blockable and Unblockable Non-Colliders).

Let $G = (J, V, E, L)$ be a CDMG and π a walk in G :

$$\pi = (v_0 \mathrel{\ast\ast} \dots \mathrel{\ast\ast} v_n).$$

We call a non-collider v_k on π an *unblockable non-collider* on π if it is not an end-node ($k \notin \{0, n\}$) and it only has outgoing edges on π to nodes in the *same* strongly connected component of G . That is, it is one of the following patterns:

- *left chain*: $v_{k-1} \leftarrow v_k \leftarrow \ast v_{k+1}$ with $v_{k-1} \in \text{Sc}^G(v_k)$;
- *right chain*: $v_{k-1} \ast \rightarrow v_k \rightarrow v_{k+1}$ with $v_{k+1} \in \text{Sc}^G(v_k)$;
- *fork*: $v_{k-1} \leftarrow v_k \rightarrow v_{k+1}$ with $v_{k-1} \in \text{Sc}^G(v_k) \wedge v_{k+1} \in \text{Sc}^G(v_k)$.

Otherwise, v_k is called a *blockable non-collider* on π .

DEFINITION 3.12 (σ -Blocked Walks).

Let $G = (J, V, E, L)$ be a CDMG and $C \subseteq J \cup V$ a subset of nodes and π a walk in G :

$$\pi = (v_0 \mathrel{\ast\ast} \dots \mathrel{\ast\ast} v_n).$$

We say that the walk π is:

- (i) *C- σ -open* (or *σ -open given C*) if and only if:
 - all colliders v_k on π are in $\text{Anc}^G(C)$;
 - all blockable non-colliders v_k on π are not in C ;

(ii) C - σ -blocked (or σ -blocked given C) if and only if at least one of the following holds:

- there exists a collider v_k on π that is *not* in $\text{Anc}^G(C)$;
- there exists a blockable non-collider v_k on π in C .

Remark. Unblockable non-colliders are always C - σ -open, regardless of the subset $C \subseteq V \cup J$.

DEFINITION 3.13 (σ -Separation).

Let $G = (J, V, E, L)$ be a CDMG and $A, B, C \subseteq J \cup V$ (not necessarily disjoint) subset of nodes. We then say that A is σ -separated from B given C in G , in symbols:

$$A \perp_G^\sigma B \mid C,$$

if every walk from a node in A to a node in $J \cup B$ is C - σ -blocked by C .¹ If that property does not hold we will write:

$$A \not\perp_G^\sigma B \mid C.$$

Remark. We also define the special case:

$$A \perp_G^\sigma B \iff A \perp_G^\sigma B \mid \emptyset.$$

PROPOSITION 3.14.

Let $G = (J, V, E, L)$ be a CDMG. For $C \subseteq J \cup V$, and $w_1, w_2 \in J \cup V$, the following are equivalent:

- (i) there exists a C - σ -open path between w_1 and w_2 in G ;
- (ii) there exists a C - σ -open walk between w_1 and w_2 in G ;
- (iii) there exists a C - σ -open walk between w_1 and w_2 in G such that all its colliders lie in C (and not just in $\text{Anc}^G(C)$).

Remark. (i) We have that $A \perp_G^\sigma B \mid C$ is equivalent to either of the following:

- every walk from a node in A to a node in $J \cup B$ is C - σ -blocked by C ;
- every path from a node in A to a node in $J \cup B$ is C - σ -blocked by C .

(ii) It also shows that if $A \not\perp_G^\sigma B \mid C$ holds then:

- there exists a (shortest) C - σ -open path from a node in A to a node in $J \cup B$;
- there exists a (shortest) C - σ -open walk from a node in A to a node in $J \cup B$ such that all its colliders lie in C .

We would like to only check all finitely many paths (non-repeated nodes) rather than the infinite number of walks (repeated nodes allowed). This can be done by slightly adjusting the definition of σ -separation.

If a CDMG G is acyclic then all non-colliders are blockable. If we leave out the condition “blockable”, in the acyclic case we can simplify the notion of σ -separation, which is usually referred to as d -separation. If G is non-acyclic then we would like to change the graph to an

¹ The choice to include J here in this place is non-standard in the literature. However, if we include J in this definition here the implied (asymmetric) separoid rules for d -/ σ -separation will be of the same form as those for Markov kernels regarding conditional independence. This is the reason we include J here.

acyclic one such that we can use the simpler notion of d -separation. This can be done with the notion of acyclification.

SECTION 3.2

| d -Separation

DEFINITION 3.15 (d -Open Walks).

Let $G = (J, V, E, L)$ be a CDMG and $C \subseteq J \cup V$ a subset of nodes and π a walk in G :

$$\pi = (v_0 ** \cdots ** v_n).$$

We say that the walk π is C - d -open if

- (i) all non-colliders on π are not in C ;
- (ii) all colliders on π are in $\text{Anc}^G(C)$. Otherwise, the walk π is called C - d -blocked.

DEFINITION 3.16 (d -Separation).

Let $G = (J, V, E, L)$ be a CDMG and $A, B, C \subseteq J \cup V$ (not necessarily disjoint) subset of nodes.

We then say that A is d -separated from B given C in G , in symbols:

$$A \perp_G^d B \mid C,$$

if every walk from a node in A to a node in $J \cup B$ is C - d -blocked by C . If that property does not hold we will write:

$$A \not\perp_G^d B \mid C.$$

Remark. This is equivalent to requiring the same only for every path π .

SECTION 3.3

| Separoid Axioms for σ -/ d -Separation

THEOREM 3.17 ((Asymmetric) Separoid Axioms for σ -/ d -Separation).

Let $G = (J, V, E, L)$ be a CDMG and $A, B, C, D \subseteq J \cup V$ subsets of nodes. Then the ternary relations $\perp = \perp_G^d$ and $\perp = \perp_G^\sigma$ satisfy the following rules.

(i) *Extended Left Redundancy:*

$$D \subseteq A \implies D \perp B \mid A.$$

(ii) *J-Restricted Right Redundancy:*

$$A \perp \emptyset \mid C \cup J$$

always holds.

(iii) *J-Inverted Right Decomposition:*

$$A \perp B \mid C \implies A \perp J \cup B \mid C.$$

(iv) *Left Decomposition:*

$$A \cup D \perp B \mid C \implies D \perp B \mid C.$$

(v) *Right Decomposition:*

$$A \perp B \cup D \mid C \implies A \perp D \mid C.$$

(vi) *Left Weak Union:*

$$A \cup D \perp B \mid C \implies A \perp B \mid D \cup C.$$

(vii) *Right Weak Union:*

$$A \perp B \cup D \mid C \implies A \perp B \mid D \cup C.$$

(viii) *Left Contraction:*

$$(A \perp B \mid D \cup C) \wedge (D \perp B \mid C) \implies A \cup D \perp B \mid C.$$

(ix) *Right Contraction:*

$$(A \perp B \mid D \cup C) \wedge (A \perp D \mid C) \implies A \perp B \cup D \mid C.$$

(x) *Right Cross Contraction:*

$$(A \perp B \mid D \cup C) \wedge (D \perp A \mid C) \implies A \perp B \cup D \mid C.$$

(xi) *Flipped Left Cross Contraction:*

$$(A \perp B \mid D \cup C) \wedge (B \perp D \mid C) \implies B \perp A \cup D \mid C.$$

In particular, we have the equivalences:

$$(A \perp B \cup D \mid C) \iff (A \perp B \mid D \cup C) \wedge (A \perp D \mid C),$$

$$(A \cup D \perp B \mid C) \iff (A \perp B \mid D \cup C) \wedge (D \perp B \mid C).$$

(xii) *J-Restricted Symmetry:*

$$A \perp B \mid C \cup J \implies B \perp A \mid C \cup J.$$

For the special case of $J = \emptyset$ we have thus (unrestricted) Symmetry.

(xiii) *Left Composition:*

$$(A \perp B \mid C) \wedge (D \perp B \mid C) \implies A \cup D \perp B \mid C.$$

(xiv) *Right Composition:*

$$(A \perp B \mid C) \wedge (A \perp D \mid C) \implies A \perp B \cup D \mid C.$$

(xv) *Left Intersection:*

$$(A \perp B \mid D \cup C) \wedge (D \perp B \mid A \cup C) \implies A \cup D \perp B \mid C,$$

if $A \cap D = \emptyset$.

(xvi) *Right Intersection:*

$$(A \perp B \mid D \cup C) \wedge (A \perp D \mid B \cup C) \implies A \perp B \cup D \mid C,$$

if $B \cap D = \emptyset$.

Proof.

□

Exercises

EXERCISE 3.1 (Walks and Graph Neighbourhoods). Consider the CDMG with input nodes J, K and directed edges

$H \rightarrow I, I \rightarrow A, A \rightarrow H, A \rightarrow B, A \rightarrow D, J \rightarrow B, K \rightarrow E, B \rightarrow F, D \rightarrow F, F \rightarrow G, E \rightarrow G,$
and bidirected edges $A \leftrightarrow D, B \leftrightarrow D, A \leftrightarrow E$.

- Count length by number of edges and identify a walk/path with its reverse. List all bidirected paths of length 3; give two bidirected walks of length 3; list all collider paths of length 5; and give two collider walks of length 5.
- Find the parents, children, siblings, ancestors, descendants, strongly connected component, and district of A .

EXERCISE 3.2 (Separation with Input Nodes). Consider $A \rightarrow B \leftarrow C \rightarrow D \leftarrow E \rightarrow F$ and $H \rightarrow E$, with input nodes A, H . Decide each statement and briefly justify it:

$$C \perp^d E, \quad C \perp^d E \mid D, \quad B \perp^d D \mid \{A, H\}, \quad B \perp^d D \mid \{C, A, H\}.$$

EXERCISE 3.3 (Separation Counterexamples). (a) Construct an acyclic directed mixed graph on three non-input nodes A, B, C , with no edge between A, C , such that $A \not\perp^d C$ and $A \not\perp^d C \mid B$.

- Give a graph with input nodes for which $A \perp^d B \mid C \Rightarrow B \perp^d A \mid C$ fails, and explain the failure.
- Give a graph whose d - and σ -separations disagree, and explain why.

SOLUTION. – In the acyclic mixed graph $A \rightarrow B \rightarrow C, B \leftrightarrow C$, there is no edge between A and C , but

$$A \not\perp^d C, \quad A \not\perp^d C \mid B.$$

The chain $A \rightarrow B \rightarrow C$ is active without conditioning; the path $A \rightarrow B \leftrightarrow C$ is active given B because B is a conditioned collider.

– In the graph $A \leftarrow C \rightarrow B \leftarrow D$ with input node D ,

$$A \perp^d B \mid C, \quad B \not\perp^d A \mid C.$$

All paths from A to B or the input node D are blocked by C . In the reverse direction, the path $B \leftarrow D$ is active, so the asymmetric separation criterion fails.

– In the graph $A \rightarrow B \rightarrow C \rightarrow D \rightarrow B$,

$$A \not\perp^\sigma C \mid \{B, D\}, \quad A \perp^d C \mid \{B, D\}.$$

On $A \rightarrow B \rightarrow C$, conditioning on B does not σ -block the path because $B, C \in \text{Sc}^G(B) = \{B, C, D\}$. For d -separation, the two simple paths $A \rightarrow B \rightarrow C$ and $A \rightarrow B \leftarrow D \leftarrow C$ are blocked by B and D , respectively.

□

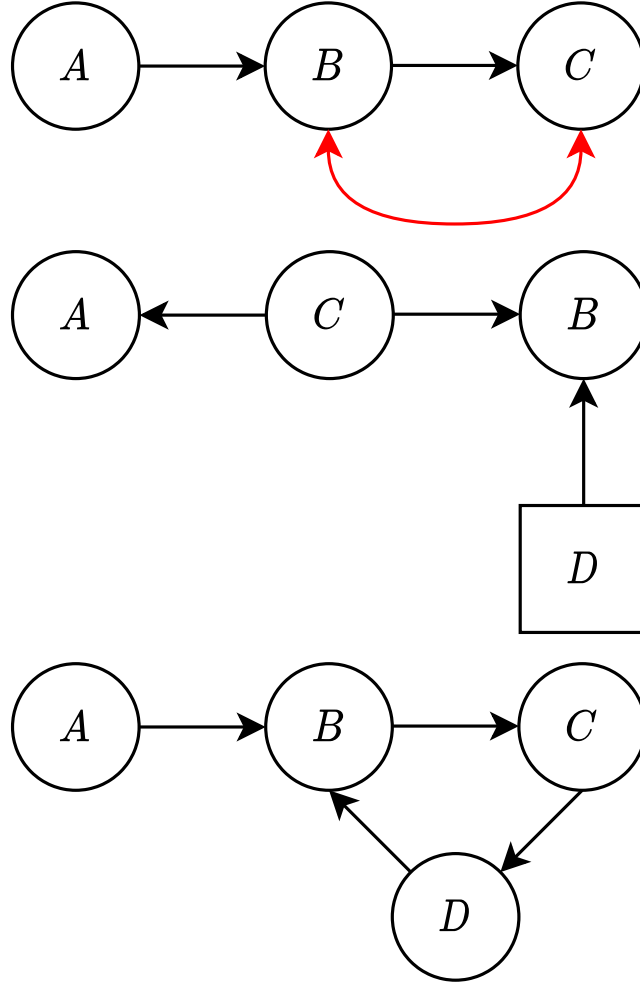


FIGURE 3.1. Separation counterexamples: mixed edges, input nodes, and a directed cycle.

EXERCISE 3.4 (Acyclification and Separation). Let G have directed edges $X_1 \rightarrow X_2 \rightarrow X_4 \rightarrow X_3 \rightarrow X_2$, $X_4 \rightarrow X_6$, and bidirected edge $X_3 \leftrightarrow X_5$.

- (a) Construct an acyclification G' . For the 2025 version, verify that the supplied graph with directed edges $X_1 \rightarrow X_i$ for $i = 2, 3, 4$, $X_2 \rightarrow X_3$, $X_2 \rightarrow X_4$, $X_4 \rightarrow X_3$, $X_4 \rightarrow X_6$, and bidirected edges $X_5 \leftrightarrow X_i$ for $i = 2, 3, 4$, is an acyclification.

- (b) Decide

$$X_1 \perp_G^\sigma X_6 \mid X_2, \quad X_1 \perp_{G'}^\sigma X_6 \mid X_2, \quad X_1 \perp_{G'}^d X_6 \mid X_2.$$

Whenever separation fails, give an active walk. Compare the results with the graph acyclification result in Proposition 3.5.2.

Causal Bayesian Networks

DEFINITION 4.1 (Causal Bayesian Network).

A *causal Bayesian network* (CBN) consists of:

- (a) a *conditional directed acyclic graph* (CDAG): $G = (J, V, E)$ with finite vertex sets, and no bi-directed edges;
- (b) a standard measurable space $\mathcal{X}_v$ for every $v \in J \cup V$;
- (c) for every $v \in V$, Markov kernels: $P_v(X_v \mid X_{\text{Pa}^G(v)})$:

$$\mathcal{X}_{\text{Pa}^G(v)} \twoheadrightarrow \mathcal{X}_v,$$

where

$$\mathcal{X}_{\text{Pa}^G(v)} = \prod_{w \in \text{Pa}^G(v)} \mathcal{X}_w.$$

DEFINITION 4.2.

Consider a causal Bayesian network with input variables with CDAG $G = (J, V, E)$ with Markov kernels $P_v(X_v \mid X_{\text{Pa}^G(v)})$ for $v \in V$. For a fixed topological ordering $<$ of G we then define the *joint Markov kernel* of the CBN:

$$X_J \twoheadrightarrow X_V$$

as follows:

$$P(X_V \mid \text{do}(X_J)) := \bigotimes_{v \in V}^> P_v(X_v \mid X_{\text{Pa}^G(v)}),$$

where the nodes v run through V in reverse ordering of $<$, i.e., all parents are on the right of all their children. And for any $A, B \subseteq J \cup V$, we also have the *marginal conditional Markov kernel*:

$$P(X_{A \cup B} \mid \text{do}(X_J)) \stackrel{\text{cond.}}{\rightsquigarrow} P(X_A \mid X_B, \text{do}(X_J)).$$

Remark. The construction is independent of the topological order $<$ by the Fubini's theorem.

DEFINITION 4.3 (CBN with Latent Variables).

A *causal Bayesian network with latent variables* (L-CBN) consists of a CBN:

$$M = \left(G^+ = (J, V^+, E^+), \left(P_v(X_v \mid X_{\text{Pa}^{G^+}(v)}) \right)_{v \in V^+} \right)$$

together with a disjoint decomposition of the output nodes $V^+ = V \dot{\cup} U$ into observed nodes $v \in V$ and *unobserved nodes* $u \in U$.

NOTATION 4.4. (i) We will also often just denote a causal Bayesian network with latent variables by the tuple:

$$M = \left(G^+ = (J, (V, U), E^+), \left(P_v \left(X_v \mid X_{\text{Pa}^{G^+}(v)} \right) \right)_{v \in V \dot{\cup} U} \right).$$

(ii) We denote the *joint Markov kernel*:

$$P(X_V, X_U \mid \text{do}(X_J)) := \bigotimes_{v \in V^+}^> P_v \left(X_v \mid X_{\text{Pa}^{G^+}(v)} \right).$$

(iii) We refer to the *marginal Markov kernel* of M :

$$P(X_V \mid \text{do}(X_J))$$

as the *observable Markov kernel*.

(iv) We call the marginalised CADMG of M :

$$G := (J, V, E, L) := (G^+)^{\setminus U}$$

the (*induced*) *observable CADMG*.

Remark.

SECTION 4.1

| Global Markov Property

THEOREM 4.5 (Global Markov Property for CBNs).

Consider a causal Bayesian network M with CADMG $G = (J, V, E, L)$ and joint Markov kernel $P(X_V \mid \text{do}(X_J))$. Then for all $A, B, C \subseteq J \cup V$ (not necessarily disjoint) we have the implication:

$$A \overset{d}{\underset{G}{\perp}} B \mid C \implies X_A \underset{P(X_V \mid \text{do}(X_J))}{\perp} X_B \mid X_C.$$

Proof.

□

THEOREM 4.6.

COROLLARY 4.7 (Global Markov Property for L-CBNs).

Proof.

□

SECTION 4.2

| Operations on Causal Bayesian Networks

DEFINITION 4.8 (Hard Interventions on CDMGs).

DEFINITION 4.9 (Node Splitting Interventions on CDMGs).

DEFINITION 4.10 (Modelling Soft Interventions on CBNs).

DEFINITION 4.11 (Modelling Hard Interventions as Soft Interventions).

Remark.

SECTION 4.3

| Standard Forms of Causal Bayesian Networks

DEFINITION 4.12 (Cliques and Maximal Cliques of Undirected Graphs).

DEFINITION 4.13 (Standard Forms of L-CBNs).

Remark.

Proof.

□

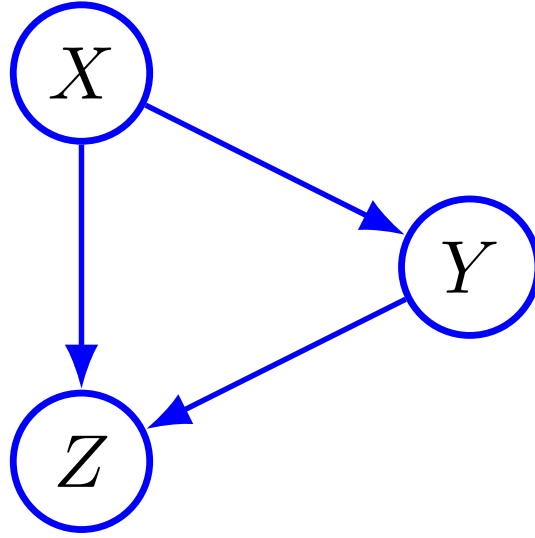


FIGURE 4.1. A causal graph with two paths from X to Z .

Exercises

EXERCISE 4.1 (Faithfulness of Gaussian CBNs). Consider the CBN of [figure 4.1](#), with $J = \emptyset$, $V = \{X, Y, Z\}$ and kernels

$$P(X) = \mathcal{N}(\mu_1, \sigma_1^2), \quad P(Y | X) = \mathcal{N}(aX + \mu_2, \sigma_2^2), \quad P(Z | X, Y) = \mathcal{N}(bY + cX + \mu_3, \sigma_3^2),$$

where all variances are positive. Faithfulness requires $X_D \perp\!\!\!\perp X_E | X_F \Rightarrow D \perp_G^d E | F$. Choose nonzero a, b, c to show that a CBN need not be faithful. *Hint. Find the joint Gaussian law and use the equivalence of uncorrelatedness and independence for Gaussian components.*

SOLUTION. Consider the graph in [figure 4.1](#), with kernels

$$\begin{aligned} P(X) &= \mathcal{N}(\mu_1, \sigma_1^2), \\ P(Y | X) &= \mathcal{N}(aX + \mu_2, \sigma_2^2), \\ P(Z | X, Y) &= \mathcal{N}(bY + cX + \mu_3, \sigma_3^2), \end{aligned}$$

where all variances are positive. Choosing $a = b = 1$ and $c = -1$ gives

$$P(Z | X) = \mathcal{N}(\mu_2 + \mu_3, \sigma_2^2 + \sigma_3^2),$$

which does not depend on X . Thus $X \perp\!\!\!\perp Z$, although the direct edge $X \rightarrow Z$ ensures $X \not\perp_G^d Z$. Hence faithfulness does not follow from the CBN assumptions.

Use independent centred Gaussian errors ε_i with variances σ_i^2 :

$$X = \mu_1 + \varepsilon_1, \quad Y = X + \mu_2 + \varepsilon_2, \quad Z = Y - X + \mu_3 + \varepsilon_3 = \mu_2 + \mu_3 + \varepsilon_2 + \varepsilon_3.$$

The last expression is independent of X . □

EXERCISE 4.2 (Reversing a Gaussian CBN). For $X \rightarrow Y$ with $p_X(x) = \mathcal{N}(x \mid 0, 1)$ and $p_Y(y \mid x) = \mathcal{N}(y \mid 5x - 3, 1)$:

- (a) Define a CBN with graph $Y \rightarrow X$ and the same joint kernel.
- (b) Find the joint interventional kernel of each CBN under $\text{do}(X = 1)$.

SOLUTION. The CBN $X \rightarrow Y$ with $X \sim \mathcal{N}(0, 1)$ and $Y \mid X = x \sim \mathcal{N}(5x - 3, 1)$ has joint distribution

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ -3 \end{pmatrix}, \begin{pmatrix} 1 & 5 \\ 5 & 26 \end{pmatrix}\right).$$

The reversed CBN $Y \rightarrow X$ has the same joint distribution when

$$Y \sim \mathcal{N}(-3, 26), \quad X \mid Y = y \sim \mathcal{N}\left(\frac{5(y + 3)}{26}, \frac{1}{26}\right).$$

Under $\text{do}(X = 1)$, their joint kernels differ:

$$\begin{aligned} P_{X \rightarrow Y}(X, Y \mid \text{do}(X = 1)) &= \delta_1(X) \otimes \mathcal{N}(Y \mid 2, 1), \\ P_{Y \rightarrow X}(X, Y \mid \text{do}(X = 1)) &= \delta_1(X) \otimes \mathcal{N}(Y \mid -3, 26). \end{aligned}$$

Writing $Y = 5X - 3 + \varepsilon$ with independent $\varepsilon \sim \mathcal{N}(0, 1)$ gives $\mathbb{E}[Y] = -3$, $\text{Var}(Y) = 26$, and $\text{Cov}(X, Y) = 5$. Gaussian conditioning gives

$$\mathbb{E}[X \mid Y = y] = \frac{5(y + 3)}{26}, \quad \text{Var}(X \mid Y = y) = 1 - \frac{25}{26} = \frac{1}{26}.$$

The intervention replaces the kernel for X and preserves the kernel for Y in each graph. $\square$

EXERCISE 4.3 (Independence in a Generative Model). Consider the generative densities and mass function

$$\begin{aligned} p_V(v) &= \mathcal{N}(v \mid 0, 1), & p_U(u) &= \text{Weibull}(u \mid 5, 3), \\ p_X(x \mid u) &= \text{Exponential}(x \mid u), & p_W(w \mid v, x) &= \text{Gamma}(w \mid v^2, x^2), \\ p_Y(y \mid v, x) &= \text{Pareto}(y \mid v^2, x^2), & p_Z(z \mid y) &= \text{Poisson}(z \mid y^2). \end{aligned}$$

Prove $V \perp\!\!\!\perp Z \mid Y$ and $V \perp\!\!\!\perp X$.

EXERCISE 4.4 (Association and Average Causal Effect). Let $J = \emptyset$, $V = \{X, Y, Z\}$, with graph $Z \rightarrow X, Z \rightarrow Y, X \rightarrow Y$ and kernels

$$\begin{aligned} P(Z) &= \text{Bernoulli}(1/3), \\ P(X \mid Z) &= \text{Bernoulli}((1 + Z)/3), \\ P(Y \mid X, Z) &= \text{Bernoulli}((1 + X + Z)/4). \end{aligned}$$

- (a) Compute $\alpha = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0]$.
- (b) Give the graph and kernels after a hard intervention on X .
- (c) Compute $\beta = \mathbb{E}[Y \mid \text{do}(X = 1)] - \mathbb{E}[Y \mid \text{do}(X = 0)]$.
- (d) If X is drug use and Y recovery, what incorrect conclusion would a doctor draw by interpreting an estimate of α from n observed patient pairs as the drug's causal effect?

EXERCISE 4.5 (Marginalisation and Interventions). Use the graph of [exercise 3.2](#), with input nodes A, H .

- Decide whether the graph is cyclic; if it is acyclic, give a topological order.
- Draw the marginal graphs obtained by eliminating C ; then C, E ; then C, E, D ; by eliminating E ; then E, C ; and by eliminating $\{C, E, D\}$ simultaneously. Check the consistency of sequential and simultaneous marginalisation.
- Draw the graph after a hard intervention on D , after adding a soft-intervention node I_B for B , and after a node-splitting intervention on E .

EXERCISE 4.6 (Monty Hall). Three doors hide one car and two goats. A contestant chooses a door; an informed host opens an unchosen goat door; the contestant then sticks or switches. Use C for initial choice, L for car location, D for opened door, F for final choice, $I = 0$ for sticking and $I = 1$ for switching, and O for final outcome.

- Define an L-CBN: draw its graph, give each variable's codomain, distinguish observed, latent and input nodes immediately before the final choice, justify the choices, and specify every output kernel.
- Compute the winning probability for each strategy. By symmetry, the initial choice may be fixed at door 1.
- Replace the informed host by an earthquake that randomly opens an unchosen door, independently of the car location. After observing a goat, draw the intervened graph and recompute both winning probabilities.
- Implement the four scenarios in Python, R, Matlab or Julia, explicitly modelling their generative processes. For the earthquake, reject samples where the opened door hides the car. Estimate each winning probability with at least 1000 samples and retain the source program.

SOLUTION. Let $C, L, D, F \in \{1, 2, 3\}$ denote the initial choice, prize location, opened door, and final choice. Let $I \in \{0, 1\}$ denote the strategy (0: stick; 1: switch), and $O \in \{\text{car}, \text{goat}\}$ the final outcome. Take C, I as input nodes, L as latent with $P(L = \ell) = 1/3$, and D, F, O as output nodes. Immediately before the final decision, C, D are known, whereas L, O are unknown; F is determined once I is specified. The graph is given in [figure 4.2](#).

For $d \neq c$, the host's kernel is

$$P(D = d \mid C = c, L = \ell) = \begin{cases} 1/2, & \ell = c, \\ 1, & \ell \neq c \text{ and } d \notin \{c, \ell\}, \\ 0, & \text{otherwise,} \end{cases}$$

and it is zero for $d = c$. The other kernels are deterministic:

$$F = \begin{cases} C, & I = 0, \\ 6 - C - D, & I = 1, \end{cases} \quad O = \begin{cases} \text{car}, & F = L, \\ \text{goat}, & F \neq L. \end{cases}$$

Consequently,

$$P(O = \text{car} \mid \text{do}(I = 0, C = c)) = \frac{1}{3}, \quad P(O = \text{car} \mid \text{do}(I = 1, C = c)) = \frac{2}{3}.$$

If an earthquake instead opens one of the two unchosen doors uniformly and independently of L , the kernel for D becomes $P(D = d \mid \text{do}(C = c)) = 1/2$ for $d \neq c$. For each fixed c , this is a mixture of hard interventions on D , whose graph is shown in [figure 4.3](#). After conditioning on the observation $L \neq D$,

$$P(O = \text{car} \mid I = i, C = c, L \neq D) = \frac{1}{2}, \quad i \in \{0, 1\}.$$

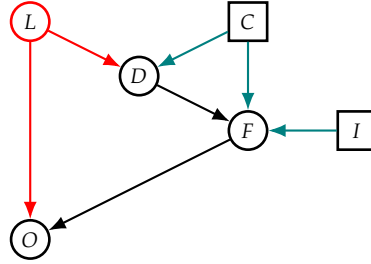


FIGURE 4.2. Monty Hall with a host who knows the prize location.

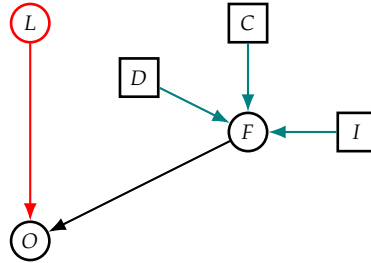


FIGURE 4.3. The graph after an independent intervention on the opened door.

With the informed host, sticking wins exactly when $L = C$; switching wins exactly when $L \neq C$. With independent random door opening, for any $d \neq c$,

$$P(L = \ell \mid C = c, D = d, L \neq D) = \frac{1}{2}, \quad \ell \neq d.$$

The two remaining doors therefore have equal winning probability. $\square$

Identification of Causal Effects in CBNs

SECTION 5.1

| Do-Calculus

THEOREM 5.1 (Do-Calculus Rules). (i) *Insertion/Deletion of Observation:* let $A, B, C \subseteq V$ and $J \subseteq D \subseteq J \cup V$ be pairwise disjoint. Assume that we want to establish the a.s. equality:

$$P(X_A \mid X_B, X_C, do(X_D)) = P(X_A \mid X_C, do(X_D)), \quad \mu_{B \cup C} - a.s.,$$

then it is sufficient to check:

$$\mu_{B \cup C} \ll P(X_B, X_C \mid do(X_D)) \ll \mu_{B \cup C},$$

and the d -separation statement:

$$A \stackrel{d}{\perp}_{G_{do(D)}} B \mid C \cup D.$$

(ii) *Action/Observation Exchange:* let $A, B, C \subseteq V$ and $J \subseteq D \subseteq J \cup V$ be pairwise disjoint. Assume that we want to establish the a.s. equality:

$$P(X_A \mid X_B, X_C, do(X_D)) = P(X_A \mid do(X_B), X_C, do(X_D)), \quad \mu_{B \cup C} - a.s.,$$

then it is sufficient to check:

$$\mu_{B \cup C} \ll P(X_B, X_C \mid do(X_D)) \ll \mu_{B \cup C}, \quad \mu_C \ll P(X_C \mid do(X_B, X_D)) \ll \mu_C,$$

and the d -separation statement:

$$A \stackrel{d}{\perp}_{G_{do(I_B, D)}} I_B \mid B \cup C \cup D.$$

(iii) *Insertion/Deletion of Action:* let $A, B, C \subseteq V$ and $J \subseteq D \subseteq J \cup V$ be pairwise disjoint. Assume that we want to establish the a.s. equality:

$$P(X_A \mid do(X_B), X_C, do(X_D)) = P(X_A \mid X_C, do(X_D)), \quad \mu_C - a.s.,$$

then it is sufficient to check:

$$\mu_C \ll P(X_C \mid do(X_D)) \ll \mu_C, \quad \mu_C \ll P(X_C \mid do(X_B, X_D)) \ll \mu_C,$$

and the d -separation statement:

$$A \stackrel{d}{\perp}_{G_{do(I_B, D)}} I_B \mid C \cup D.$$

- (iv) *Deletion of Input: let $A, B, C \subseteq V$ and $D \subseteq J \cup V$ be pairwise disjoint. Assume that we want to find a Markov kernel (lhs) with a.s. equality:*

SECTION 5.2

| Adjustment Criteria and Formulae

Exercises

EXERCISE 5.1 (Backdoor Adjustment with a Latent Cause). Consider an L-CBN with $J = \emptyset$, $V = \{X, Y, Z\}$, latent L , and augmented edges $L \rightarrow Z$, $L \rightarrow Y$, $Z \rightarrow X$, $Z \rightarrow Y$, $X \rightarrow Y$. The kernels $P(Y \mid \text{do}(X, Z, L))$, $P(X \mid \text{do}(Z))$, $P(Z \mid \text{do}(L))$ and $P(L)$ are unknown, but observational data give

$$P(Z) = \text{Bernoulli}(1/3), \quad P(Y \mid X, Z) = \text{Bernoulli}((1 + X + Z)/4).$$

Draw the graph after adding I_X , verify the backdoor criterion, and use adjustment to calculate $\mathbb{E}[Y \mid \text{do}(X = 1)] - \mathbb{E}[Y \mid \text{do}(X = 0)]$. Assume the required absolute continuity.

EXERCISE 5.2 (Backdoor Walks and the Adjustment Criterion). Let $G^+ = (J, (V, U), E^+)$ be a DAG with $J = \emptyset$ and pairwise disjoint $A, B, C \subseteq V$. In the graph obtained by adding intervention variables I_B , consider

$$(i) \quad C \perp^d I_B, \quad (ii) \quad A \perp^d I_B \mid B \cup C.$$

A backdoor walk from A to B ends in an edge $v \rightarrow b$, $b \in B$. Compare the alternative conditions

$$(i') \quad C \cap \text{Desc}^G(B) = \emptyset, \quad (ii') \quad \text{every backdoor walk from } A \text{ to } B \text{ is } d\text{-blocked by } C.$$

(a) Prove $(i) \Leftrightarrow (i')$.

(b) Prove $(ii') \Rightarrow (ii)$.

(c) Construct a DAG and disjoint A, B, C for which (ii) holds but (ii') does not.

EXERCISE 5.3 (Simpson's Paradox and Causal Models). Insurance records of patients with a deadly disease give the following counts; treatment assignment was not random and its reasons are unknown.

Population	Treatment	Recovered	Not recovered	Total
All	Drug	20	20	40
	No drug	16	24	40
	Total	36	44	80
Male	Drug	18	12	30
	No drug	7	3	10
	Total	25	15	40
Female	Drug	2	8	10
	No drug	9	21	30
	Total	11	29	40

(a) Compute the aggregate recovery rates for drug and no drug.

(b) Based on those rates, would you take the drug if diseased?

(c) Compute the treatment-specific recovery rates within each sex.

(d) Would this change your treatment choice?

(e) What would you advise a patient of unknown sex?

Let D denote drug use and R recovery. Consider three causal graphs:

- (i) $M \rightarrow D, M \rightarrow R, D \rightarrow R;$
- (ii) $D \rightarrow M, M \rightarrow R, D \rightarrow R;$
- (iii) $L_1 \rightarrow D, L_1 \rightarrow M, L_2 \rightarrow M, L_2 \rightarrow R, D \rightarrow R,$

where L_1, L_2 are latent. In (i), M is sex; in (ii), it may instead be blood pressure.

(f) For (i), apply backdoor adjustment to express $P(R \mid \text{do}(D))$ using observed distributions.

(g) Is it equal to $P(R \mid D)$?

(h) Advise a patient with unknown M under model (i).

(i) For (ii), express $P(R \mid \text{do}(D))$ by backdoor adjustment.

(j) Is it equal to $P(R \mid D)$?

(k) Advise a patient with unknown M under model (ii).

(l) Interpret M, L_1, L_2 in model (iii).

(m) Express $P(R \mid \text{do}(D))$ using observed quantities in model (iii).

(n) Is it equal to $P(R \mid D)$?

(o) Advise a patient with unknown M under model (iii).

EXERCISE 5.4 (Recovering from Selection Bias). Use the graph of [figure 5.1](#), with latent U and binary selection variable S . The goal is to identify $P(Y \mid \text{do}(X))$ using only $P(W_1, W_2, X, Y \mid S = 1)$; the final expression may contain neither an intervention nor U . Assume all absolute continuities required by do-calculus.

- (a) Prove $P(Y \mid \text{do}(X)) = P(Y \mid S, \text{do}(X))$.
- (b) Prove $P(Y \mid S, \text{do}(X)) = P(Y \mid S, W_2, \text{do}(X)) \circ P(W_2 \mid S)$.
- (c) Prove $P(Y \mid S, W_2, \text{do}(X)) = P(Y \mid S, W_2, X)$.
- (d) Explain why these identities solve the identification problem.

SOLUTION. Consider the graph in [figure 5.1](#), with latent U and binary selection variable S . Only $P(W_1, W_2, X, Y \mid S = 1)$ is observed. Under the absolute continuity conditions required by do-calculus,

$$P(Y \mid \text{do}(X)) = P(Y \mid X, W_2, S = 1) \circ P(W_2 \mid S = 1).$$

Thus the causal effect is identifiable from the selected observational distribution.

After deleting incoming edges to X , the path $S \leftarrow W_1 \rightarrow W_2 \leftarrow U \rightarrow Y$ is blocked at the collider W_2 . Hence $Y \perp^d S \mid X$ in this graph, and observation insertion gives

$$P(Y \mid \text{do}(X)) = P(Y \mid S, \text{do}(X)).$$

The chain rule, followed by removal of the intervention on X from the non-descendant kernel, gives

$$P(Y \mid S, \text{do}(X)) = P(Y \mid W_2, S, \text{do}(X)) \circ P(W_2 \mid S, \text{do}(X))$$

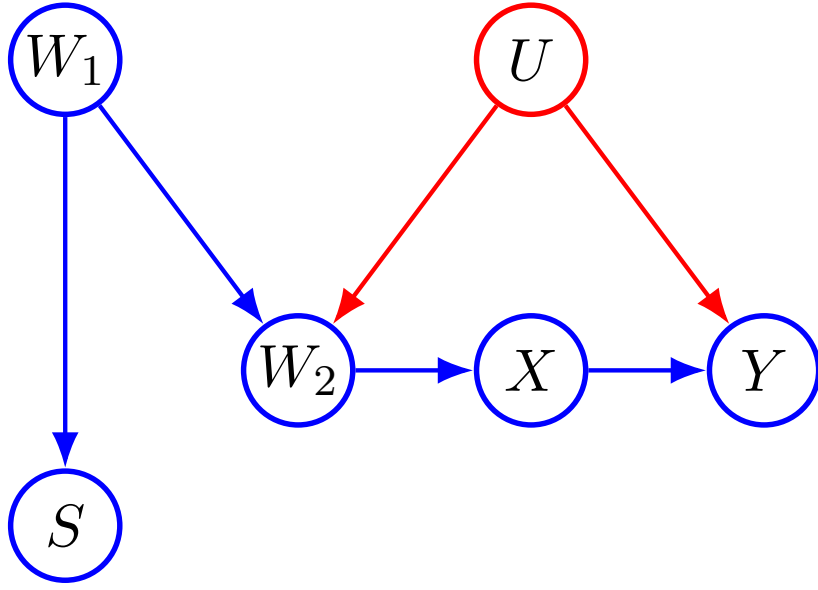


FIGURE 5.1. Selection through W_1 with an unobserved common cause U of W_2 and Y .

$$= P(Y \mid W_2, S, \text{do}(X)) \circ P(W_2 \mid S).$$

After deleting outgoing edges from X , the path $X \leftarrow W_2 \leftarrow U \rightarrow Y$ is blocked by W_2 . Thus $Y \perp^d X \mid \{W_2, S\}$ in that graph; action/observation exchange yields

$$P(Y \mid W_2, S, \text{do}(X)) = P(Y \mid W_2, S, X).$$

Evaluating at $S = 1$ proves the formula. Both factors are obtained by marginalisation and conditioning of the observed selected distribution. $\square$

Structural Causal Models

DEFINITION 6.1 (Structural Causal Model).

A structural causal model M is a tuple $(J, V, W, \mathcal{X}, P, f)$ such that

- (i) the *exogenous input variables*, the *endogenous variables* and the *exogenous random variables* are indexed by disjoint (finite) index sets J, V, W , respectively;
- (ii) the domain $\mathcal{X} = \prod_{i \in J \cup V \cup W} \mathcal{X}_i$ is a product of standard measurable spaces $\mathcal{X}_i$;
- (iii) the *exogenous distribution* P is a probability distribution on $\mathcal{X}_W$ that factorises as a product $P = \bigotimes_{w \in W} P_w$ of probability distributions $P_w \in \mathcal{P}(\mathcal{X}_w)$, one for each exogenous space $\mathcal{X}_w$;
- (iv) the *causal mechanism* is specified by the measurable function $f : \mathcal{X} \rightarrow \mathcal{X}_V$.

Remark. (i) If f and P depend in a measurable way on exogenous parameters $\theta \in \Theta$, we write $M^\theta = (J, V, W, \mathcal{X}, P^\theta, f^\theta)$.

- (ii) Our definition of SCMs is *unorthodox*, as we allow for exogenous input variables, and cyclic causal dependencies between the endogenous variables.

There are three crucial assumptions embodied in the modeling approach using SCMs:

- (i) An SCM models a system with (free or random) inputs and outputs, such that the inputs are *not caused by* the outputs (the inputs are ‘exogenous’ w.r.t. the combined in- and outputs).
- (ii) Exogenous random variables are *mutually independent*.
- (iii) The distribution of the exogenous random variables is *independent* of the values of the exogenous input variables.

SECTION 6.1

| Solving the Structural Equations

DEFINITION 6.2.

Let $M = (J, V, W, \mathcal{X}, P, f)$ be an SCM and $x_J \in \mathcal{X}_J$ an input value. A random variable $X_{V \cup W}^{\text{do}(x_J)}$ with codomain $\mathcal{X}_V \times \mathcal{X}_W$ is called a *potential outcome of M for input x_J* if the following two conditions hold:

- (i) its W -component has the exogenous distribution specified by M : $X_W^{\text{do}(x_J)} \sim P$;

(ii) it satisfies the *structural equations entailed by M for input x_J* :

$$X_V^{\text{do}(x_J)} = f\left(x_J, X_V^{\text{do}(x_J)}, X_W^{\text{do}(x_J)}\right), \quad a.s.$$

Remark. In case $J = \emptyset$ we also write $X_{V \cup W} = X_{V \cup W}^{\text{do}(\ast)}$ and refer to it simply as an outcome of M . A common way to write potential outcomes is as $X_{V \cup W}(x_J)$.

EXAMPLE 6.3.

EXAMPLE 6.4 (Linear Regression Model).

DEFINITION 6.5.

Let $M = (J, V, W, \mathcal{X}, P, f)$ be an SCM. Let $(\mathcal{U} \times \mathcal{X}_J, K(U | X_J))$ be a transition probability space and $X : \mathcal{U} \times \mathcal{X}_J \rightarrow \mathcal{X}$ be a conditional random variable. If its push-forward Markov kernel $K(X | X_J)$ satisfies

$$K(X_W | X_J) = P(X_W), \quad K(X_J | X_J) = \delta(X_J | X_J)$$

and

$$X_V = f(X_J, X_V, X_W), \quad K(X | X_J)\text{-a.s.},$$

then X (and $X_{V \cup W}$) is called a *solution of M* .

Remark. If $J = \emptyset$, a solution of an SCM can be identified with a random variable $X_{V \cup W}$ with codomain $\mathcal{X}_V \times \mathcal{X}_W$ such that $X_W \sim P$ and that satisfies the structural equations: $X_v = f_v(X_V, X_W)$ a.s. for each $v \in V$.

Let $M = (J, V, W, \mathcal{X}, P, f)$ be an SCM and $X : \mathcal{U} \times \mathcal{X}_J \rightarrow \mathcal{X}_{V \cup W}$ be a solution of M . Then for any $x_J \in \mathcal{X}_J$,

$$\begin{aligned} X_{V \cup W}^{\text{do}(x_J)} : \mathcal{U} &\rightarrow \mathcal{X}_{V \cup W} \\ u &\mapsto X(u, x_J) \end{aligned}$$

is a (potential) outcome of M (for input x_J). Hence, we can think of a solution as a measurable family of potential outcomes that share an underlying probability space.

EXAMPLE 6.6.

NOTATION 6.7.

Remark.

We can construct solutions and (potential) outcomes in terms of solution functions of an SCM:

DEFINITION 6.8.

Given an SCM $M = (J, V, W, \mathcal{X}, P, f)$, we call a measurable function $g : \mathcal{X}_J \times \mathcal{X}_W \rightarrow \mathcal{X}_V$ a solution function of M if for all $x_J \in \mathcal{X}_J$, and for all $x_W \in \mathcal{X}_W$, $g(x_J, x_W)$ satisfies the structural equations:

$$g(x_J, x_W) = f(x_J, g(x_J, x_W), x_W).$$

SECTION 6.2

| Equivalence Notions

DEFINITION 6.9 (Special Case).

Let $M = (J, V, W, \mathcal{X}, P, f)$ and $\tilde{M} = (\tilde{J}, \tilde{V}, \tilde{W}, \tilde{\mathcal{X}}, \tilde{P}, \tilde{f})$ be two simple SCMs with $J = \tilde{J} = \emptyset$ and $O \subseteq V \cap \tilde{V}$ a common subset. We say that:

- (i) M and $\tilde{M}$ are *observably equivalent w.r.t. O* if $\mathcal{X}_O = \tilde{\mathcal{X}}_O$ and their marginal distributions coincide: $P_M(X_O) = P_{\tilde{M}}(X_O)$;
- (ii)

DEFINITION 6.10.

Let $M = (J, V, W, \mathcal{X}, P, f)$ and $\tilde{M} = (\tilde{J}, \tilde{V}, \tilde{W}, \tilde{\mathcal{X}}, \tilde{P}, \tilde{f})$ be two simple SCMs and $O \subseteq V \cap \tilde{V}$ a common subset. We say that:

- (i) M and $\tilde{M}$ are *observably equivalent w.r.t. O* if $\mathcal{X}_O = \tilde{\mathcal{X}}_O$ and their marginal Markov kernels coincide:

$$P_M(X_O \mid \text{do}(X_J)) = P_{\tilde{M}}(X_O \mid \text{do}(X_{\tilde{J}})).$$

This has to be interpreted as both Markov kernels being a version of a Markov kernel $\mathcal{X}_{J \cap \tilde{J}} \rightarrow \mathcal{X}_O$, i.e., $P_M(X_O \mid \text{do}(X_J))$ must be essentially constant in $x_{J \cap \tilde{J}}$ and $P_{\tilde{M}}(X_O \mid \text{do}(X_{\tilde{J}}))$ must be essentially constant in $x_{\tilde{J} \cap J}$;

- (ii) M and $\tilde{M}$ are *interventionally equivalent w.r.t. O* if for every subset $T \subseteq O$ the intervened SCMs $M_{\text{do}(T)}$ and $\tilde{M}_{\text{do}(T)}$ are observably equivalent w.r.t. $O \setminus T$.

SECTION 6.3

| Interventions

DEFINITION 6.11.

Let $M = (J, V, W, \mathcal{X}, P, f)$ be an SCM and $T \subseteq J \cup V$ the *intervention target*. We define three variants of *hard interventions*:

- (i) for a *specified target value* $\xi_T \in \mathcal{X}_T$, define

$$M_{\text{do}(X_T=\xi_T)} = (J \setminus T, V \cup T, W, \mathcal{X}, P, (f_{V \setminus T}, \xi_T));$$

- (ii) for a *stochastic target value* with distribution $Q_T \in \bigotimes_{t \in T} \mathcal{P}(\mathcal{X}_t)$, define

$$M_{\text{do}(X_T \sim Q_T)} = (J \setminus T, V \setminus T, W \cup T, \mathcal{X}, P \otimes Q_T, f_{V \setminus T});$$

- (iii) for an *unspecified target value*, define

$$M_{\text{do}(T)} = (J \cup T, V \setminus T, W, \mathcal{X}, P, f_{V \setminus T}).$$

SECTION 6.4

| Unique Solvability and Simple SCMs

PROPOSITION 6.12.

Let $M = (J, V, W, \mathcal{X}, P, f)$ be a linear SCM. Then, for any set $L \subseteq V$, the submodel $M^{[L]}$ is uniquely solvable if and only if the matrices $I_L - B_{LL}(x_J, x_W)$ are invertible for all $x_J \in \mathcal{X}_J, x_W \in \mathcal{X}_W$, where $I_L \in \mathbb{R}^{L \times L}$ is the identity matrix and $B_{LL}(x_J, x_W) \in \mathbb{R}^{L \times L}$ the submatrix of $B(x_J, x_W)$. Its unique solution function is then:

$$g : \mathbb{R}^{V \setminus L} \times \mathbb{R}^{J \cup W} \rightarrow \mathbb{R}^L$$

$$: (x_{V \setminus L}, x_J, x_W) \mapsto (I_L - B_{LL}(x_J, x_W))^{-1} (B_{L, V \setminus L}(x_J, x_W) x_{V \setminus L} + c_L(x_J, x_W)).$$

DEFINITION 6.13.

An SCM $M = (J, V, W, \mathcal{X}, P, f)$ is called *uniquely solvable* if it has a unique solution function.

Remark. This means two things:

(i) it has a solution function $g : \mathcal{X}_J \times \mathcal{X}_W \rightarrow \mathcal{X}_V$, i.e., a measurable function that satisfies

$$\forall x_J \in \mathcal{X}_J \forall x_W \in \mathcal{X}_W : g(x_J, x_W) = f(x_J, g(x_J, x_W), x_W);$$

(ii) all its solution functions must be equal, i.e., for any solution functions $g, \tilde{g} : \mathcal{X}_J \times \mathcal{X}_W \rightarrow \mathcal{X}_V$, we have that for all $x_J \in \mathcal{X}_J$, for all $x_W \in \mathcal{X}_W$:

$$g(x_J, x_W) = \tilde{g}(x_J, x_W).$$

THEOREM 6.14.

DEFINITION 6.15.

PROPOSITION 6.16.

DEFINITION 6.17.

An SCM $M = (J, V, W, \mathcal{X}, P, f)$ is called *simple* if the intervened SCM $M_{\text{do}(T)}$ is uniquely solvable for all $T \subseteq V$.

Remark. This includes unique solvability of M itself for $T = \emptyset$.

COROLLARY 6.18.

If M is simple, then its Markov kernel $P_M(X_V, X_W \mid \text{do}(X_J))$ and all intervened Markov kernels

$$P_{M_{\text{do}(T)}}(X_{V \setminus T}, X_W \mid \text{do}(X_{J \cup T}))$$

for $T \subseteq V$ exist and are unique.

NOTATION 6.19.

EXAMPLE 6.20.

SECTION 6.5

| Marginalisation

SECTION 6.6

| Acyclifications

We also define an acyclification operation on simple SCMs.

DEFINITION 6.21.

Let $M = (J, V, W, \mathcal{X}, P, f)$ be a simple SCM with causal graph $G(M)$. For each strongly connected component $C \subseteq V$ of $G(M)$ (i.e., a set of the form $\text{Sc}^{G(M)}(v)$ for $v \in V$), let $g^{[C]} : \mathcal{X}_{J \cup (V \setminus C) \cup W} \rightarrow \mathcal{X}_C$ be the unique solution function for $M^{[C]}$. Define $\tilde{f} : \mathcal{X}_J \times \mathcal{X}_V \times \mathcal{X}_W \rightarrow \mathcal{X}_V$ by its components

$$\tilde{f}_v(x_J, x_V, x_W) = g_v^{[\text{Sc}^{G(M)}(v)]}(x_J, x_{V \setminus \text{Sc}^{G(M)}(v)}, x_W)$$

for $v \in V$. $M^{\text{acy}} = (J, V, W, \mathcal{X}, P, \tilde{f})$ is called the *acyclification* of M .

| Exercises

EXERCISE 6.1 (A Gaussian CBN as an SCM). Use the Gaussian CBN in [exercise 4.1](#).

- (a) Define it as an iSCM.
- (b) Find its solution function $g : \mathcal{X}_J \times \mathcal{X}_W \rightarrow \mathcal{X}_V$.
- (c) Choose nonzero a, b, c giving $X \perp\!\!\!\perp Z$, so that the SCM is not faithful.
- (d) For each $y \in \mathbb{R}$, specify $M_{\text{do}(Y=y)}$.
- (e) Compute $\mathbb{E}[Z \mid \text{do}(Y = y)]$.
- (f) Compute $\mathbb{E}[Z \mid \text{do}(Y = y + 1)] - \mathbb{E}[Z \mid \text{do}(Y = y)]$.

EXERCISE 6.2 (Intervening on an Exogenous Variable). Consider $C = \alpha + \beta W$, $N = \gamma + \delta W$, where wealth W is an exogenous random variable with distribution $P(W)$ and chocolate consumption C and Nobel-prize count N are endogenous. Given a solution with law $P(C, N, W)$, consider $\text{do}(C = c)$, $\text{do}(N = n)$ and $\text{do}(W = w)$. For each intervention, determine which distributions $P(X_i \mid \text{do}(X_j = x_j))$, $i \neq j$, $i, j \in \{C, N, W\}$ remain unchanged and which may change. Compare this with intuitive causal effects. Assess whether the equations are realistic and how they could be improved.

EXERCISE 6.3 (Wealth, Chocolate and Nobel Prizes). Consider $W = \varepsilon_W$, $C = \alpha + \beta W$, $N = \gamma + \delta W$, with exogenous $\varepsilon_W \sim P(\varepsilon_W)$ and endogenous W, C, N . For $\text{do}(C = c)$, $\text{do}(N = n)$ and $\text{do}(W = w)$, determine which non-target distributions are unchanged and which may change. Compare the result with intuitive causal effects and discuss the realism of the equations and possible improvements.

EXERCISE 6.4 (Merging Variables). Show that the merged-variable model in Example 6.1.1 can be represented by a simple iSCM. Is the class of iSCMs generally closed under merging variables?

EXERCISE 6.5 (Mixtures of SCMs). Show that the mixture model in Example 6.1.2 can be represented by a simple iSCM. Is the class of iSCMs generally closed under taking mixtures?

EXERCISE 6.6 (The Half-Adder). The half-adder in [figure 6.1](#) has binary input bits A, B , sum bit S and carry bit C . Initially, A, B are exogenous input variables without a probability distribution.

- (a) Formally specify a deterministic SCM $M = (J, V, W, \mathcal{X}, P, f)$.
- (b) Find a solution function $g : \mathcal{X}_J \times \mathcal{X}_W \rightarrow \mathcal{X}_V$ and prove its uniqueness.
- (c) Give the sum and carry for every input pair.

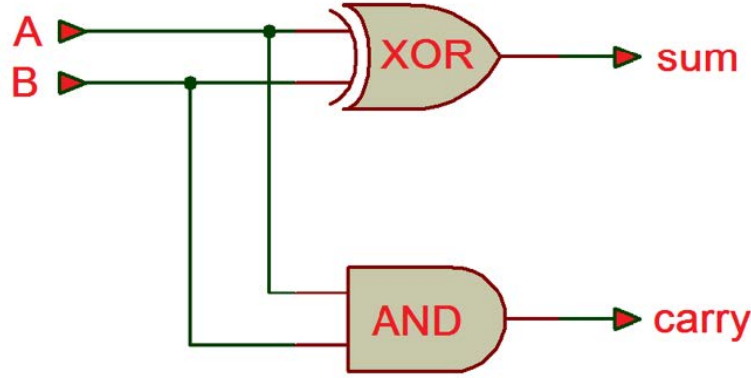


FIGURE 6.1. A half-adder computes the sum bit by XOR and the carry bit by AND.

- (d) Model hard interventions on A and S , and a soft intervention replacing XOR by OR. Give the possible outputs of each intervened solution and compare them with the original outputs.

Now let A, B be independent fair Bernoulli random inputs. In the 2024 formulation, they are exogenous random variables; in the 2025 formulation, they are endogenous copies of independent exogenous random variables.

- (e) Specify the SCM M' for each formulation.
- (f) Find the observational law of $(S, C) \in \{0, 1\}^2$ and name the measure-theoretic operation used.
- (g) Find the law of (S, C) under $\text{do}(S = 1)$.
- (h) Find the law of (S, C) under $\text{do}(A = 0)$.

SOLUTION. A deterministic half-adder has input nodes $J = \{A, B\}$, output nodes $V = \{S, C\}$, no random nodes ($W = \emptyset$), and binary spaces. The exogenous distribution is the point mass on the singleton $\mathcal{X}_\emptyset$. Its mechanisms and unique solution function are

$$f_S(a, b, s, c) = a \oplus b, \quad f_C(a, b, s, c) = ab, \quad g(a, b) = (a \oplus b, ab),$$

where $\oplus$ denotes XOR. The truth table is

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

and $\text{im}(g) = \{(0, 0), (1, 0), (0, 1)\}$. Since neither mechanism depends on s, c , every solution function must equal g . Hard interventions on an input and on the sum give

$$g^{\text{do}(A=a)}(b) = (a \oplus b, ab), \quad g^{\text{do}(S=1)}(a, b) = (1, ab).$$

The possible outputs under $\text{do}(A = 0)$ are $\{(0, 0), (1, 0)\}$; under $\text{do}(A = 1)$ they are $\{(1, 0), (0, 1)\}$; and under $\text{do}(S = 1)$ they are $\{(1, 0), (1, 1)\}$. Replacing XOR by OR is a soft intervention with

$$\tilde{g}(a, b) = (a \vee b, ab), \quad \text{im}(\tilde{g}) = \{(0, 0), (1, 0), (1, 1)\}.$$

For random inputs, take $J = \emptyset$, $V = \{A, B, S, C\}$, $W = \{A', B'\}$, binary spaces, and

$$P = P_{A'} \otimes P_{B'}, \quad A', B' \sim \text{Bernoulli}(1/2),$$

with $f_A = \varepsilon_{A'}$, $f_B = \varepsilon_{B'}$, $f_S = A \oplus B$, and $f_C = AB$. The solution is

$$g'(\varepsilon_{A'}, \varepsilon_{B'}) = (\varepsilon_{A'}, \varepsilon_{B'}, \varepsilon_{A'} \oplus \varepsilon_{B'}, \varepsilon_{A'} \varepsilon_{B'}).$$

The observational and interventional laws are push-forwards of P :

s	c	$P(S = s, C = c)$	$P(S = s, C = c \mid \text{do}(S = 1))$	$P(S = s, C = c \mid \text{do}(A = 0))$
0	0	1/4	0	1/2
1	0	1/2	3/4	1/2
0	1	1/4	0	0
1	1	0	1/4	0

□

EXERCISE 6.7 (Strict Graph Inclusion after Marginalisation). Construct a simple SCM M and $L \subset V$ for which

$$G(M_{\setminus L}) \subsetneq G(M)^{\setminus L}.$$

Specify the marginal SCM and draw both graphs. The Gaussian model of [exercise 6.1](#) may provide inspiration.

EXERCISE 6.8 (Equivalence of SCMs). (a) Prove that SCM equivalence is an equivalence relation.

(b) Prove that equivalence is preserved by each of the three types of hard intervention, for applicable targets in $J \cup V \cup W$.

(c) Prove that equivalent SCMs have the same solution functions. Deduce, using intervention invariance, that unique solvability and simplicity are invariants of equivalence.

(d) Prove that equivalent SCMs have the same potential outcomes, solutions, and Markov kernels.

EXERCISE 6.9 (Invariants of SCM Equivalence). Let $M = (J, V, W, \mathcal{X}, P, f)$ and $\tilde{M} = (J, V, W, \mathcal{X}, \tilde{P}, \tilde{f})$ satisfy $M \equiv \tilde{M}$. Prove the following claims.

(a) Equivalence is preserved by all three hard-intervention types with targets in $V \cup J$; by soft interventions $T \leftarrow \hat{f}_T$, for $T \subseteq V$ and measurable $\hat{f}_T : \mathcal{X}_J \times \mathcal{X}_W \times \mathcal{X}_V \rightarrow \mathcal{X}_T$; and by adding intervention variables I_B for $B \subseteq V \cup J$.

(b) Equivalent iSCMs have equivalent submodels, and composition preserves equivalence.

(c) The models have the same null sets, so M -almost surely and $\tilde{M}$ -almost surely have the same meaning.

(d) The models have the same potential outcomes, solutions, Markov kernels, and partial solution functions.

- (e) For every $L \subseteq V$, partial solvability, essential unique solvability, essential uniqueness of a given partial solution function, and simplicity hold for M if and only if they hold for $\bar{M}$.
- (f) Essential linearity is preserved.

See Proposition 6.7.2.

EXERCISE 6.10 (Marginalising a Linear SCM). Let $J = \emptyset$, $V = \{A, B, C, D\}$, $W = \{A', B', C', D'\}$, $\mathcal{X} = \mathbb{R}^8$, and $P = \bigotimes_{i \in V} \mathcal{N}(\mu_i, \sigma_i^2)$ on the corresponding errors. For nonzero coefficients, the equations are

$$A = \varepsilon_{A'}, \quad B = \alpha_{ab}A + \alpha_{db}D + \varepsilon_{B'}, \quad C = \alpha_{bc}B + \alpha_{dc}D + \varepsilon_{C'}, \quad D = \alpha_{ad}A + \varepsilon_{D'}.$$

- (a) Construct a solution function.
- (b) Draw $G(M)$.
- (c) For latent $L = \{D\}$, specify $M_{\setminus L}$ and its solution function.
- (d) Prove $G(M_{\setminus L}) \subseteq G(M)^{\setminus L}$.
- (e) Prove observational equivalence on $V \setminus L$ directly from the solution functions and kernels.
- (f) Prove observational equivalence of $M_{\text{do}(B)}$ and $(M_{\setminus L})_{\text{do}(B)}$ on $\{A, C\}$.

SOLUTION. Let $J = \emptyset$, $V = \{A, B, C, D\}$, $W = \{A', B', C', D'\}$, and $\mathcal{X} = \mathbb{R}^8$. Let P be the product of independent Gaussian error laws $\varepsilon_{i'} \sim \mathcal{N}(\mu_{i'}, \sigma_{i'}^2)$. With nonzero coefficients, the structural equations are

$$\begin{aligned} A &= \varepsilon_{A'}, & D &= \alpha_{ad}A + \varepsilon_{D'}, \\ B &= \alpha_{ab}A + \alpha_{db}D + \varepsilon_{B'}, & C &= \alpha_{bc}B + \alpha_{dc}D + \varepsilon_{C'}. \end{aligned}$$

The causal graph is shown in [figure 6.2](#). Put $k = \alpha_{ab} + \alpha_{db}\alpha_{ad}$. The unique solution is

$$\begin{aligned} g_A &= \varepsilon_{A'}, \\ g_D &= \alpha_{ad}\varepsilon_{A'} + \varepsilon_{D'}, \\ g_B &= k\varepsilon_{A'} + \varepsilon_{B'} + \alpha_{db}\varepsilon_{D'}, \\ g_C &= (\alpha_{bc}k + \alpha_{dc}\alpha_{ad})\varepsilon_{A'} + \alpha_{bc}\varepsilon_{B'} + \varepsilon_{C'} + (\alpha_{bc}\alpha_{db} + \alpha_{dc})\varepsilon_{D'}. \end{aligned}$$

Marginalising $L = \{D\}$ retains the same error variables and distribution, with $V' = \{A, B, C\}$, $\mathcal{X}' = \mathbb{R}^7$, and mechanisms

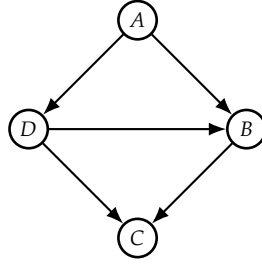
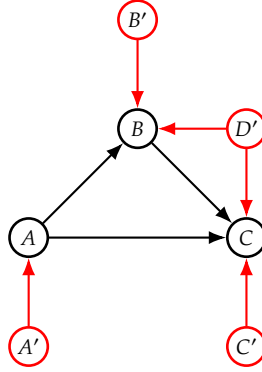
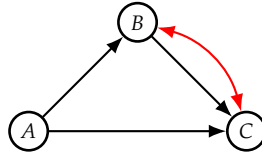
$$\begin{aligned} \tilde{f}_A &= \varepsilon_{A'}, \\ \tilde{f}_B &= kA + \varepsilon_{B'} + \alpha_{db}\varepsilon_{D'}, \\ \tilde{f}_C &= \alpha_{bc}B + \alpha_{dc}\alpha_{ad}A + \varepsilon_{C'} + \alpha_{dc}\varepsilon_{D'}. \end{aligned}$$

In particular, B remains a structural argument of $\tilde{f}_C$. The augmented marginal graph is given in [figure 6.3](#), and the latent projection in [figure 6.4](#). Its edges are

$$A \rightarrow B, \quad A \rightarrow C, \quad B \rightarrow C, \quad B \leftrightarrow C.$$

The marginal SCM has all these edges if $k \neq 0$, and loses $A \rightarrow B$ if $k = 0$. Hence

$$G(M_{\setminus L}) \subseteq G(M)^{\setminus L}.$$

FIGURE 6.2. The original causal graph $G(M)$.FIGURE 6.3. The augmented marginal graph when $k \neq 0$.FIGURE 6.4. The latent projection; the edge $A \rightarrow B$ is absent from the marginal SCM when $k = 0$.

Substituting $D = \alpha_{ad}A + \varepsilon_{D'}$ into the original mechanisms gives the displayed marginal mechanisms. Solving them yields

$$\tilde{g} = (g_A, g_B, g_C),$$

and therefore

$$P_M(A, B, C) = (g_A, g_B, g_C)_* P = \tilde{g}_* P = P_{M_{\setminus L}}(A, B, C).$$

Under $\text{do}(B = b)$, both models have the same solution on $\{A, C\}$:

$$h(b, \varepsilon) = (\varepsilon_{A'}, \alpha_{bc}b + \alpha_{dc}\alpha_{ad}\varepsilon_{A'} + \alpha_{dc}\varepsilon_{D'} + \varepsilon_{C'}).$$

Thus

$$P_{M_{\text{do}(B)}}(A, C \mid b) = h(b, \cdot)_* P = P_{(M_{\setminus L})_{\text{do}(B)}}(A, C \mid b),$$

which proves observational equivalence after this intervention. $\square$

EXERCISE 6.11 (Solvability after Marginalisation). Construct an iSCM M and $L \subseteq O \subseteq V$ such that $M_{\setminus L}$ is essentially uniquely solvable with respect to $O \setminus L$, but M is not essentially uniquely solvable with respect to O .

EXERCISE 6.12 (Simplicity of Coupled Oscillators). Consider the SCM for the equilibrium of the damped coupled harmonic oscillator in Example 6.12.1.

- (a) Put the model in linear matrix form and show essential linearity.
- (b) Prove $I_V - B_{VV}$ is invertible. Factor it as DT , with D diagonal and T tridiagonal, and use

$$T_{ii} = k_{i-1} + k_i, \quad T_{i,i+1} = T_{i+1,i} = -k_i, \quad \det T = \sum_{i=0}^d \prod_{\substack{j=0 \\ j \neq i}}^d k_j.$$

- (c) For every intervention target $Q \subseteq V$, prove $I_{V \setminus Q} - B_{V \setminus Q, V \setminus Q}$ is invertible. Conclude simplicity.

Hint. Determinants multiply under matrix products, and the determinant of a block-diagonal matrix is the product of its block determinants.

EXERCISE 6.13 (Unique Solvability of Price, Supply and Demand). Consider the dynamical system and SCM for price, supply and demand in Example 6.12.3. For each of the eight subsets $T \subseteq \{D, S, R\}$, determine whether $M_{\text{do}(T)}$ is essentially uniquely solvable. Use the linear-SCM invertibility criterion of [proposition 6.12](#).

EXERCISE 6.14 (Acyclifying a Feedback System). Let $J = \emptyset$, $V = \{A, B, C, D, E, F\}$, $W = \{A', B', C', D', E', F'\}$ and $X = \mathbb{R}^{12}$. The errors are independent Gaussians with respective means μ_i and standard deviations σ_i . The equations are

$$\begin{aligned} A &= \alpha_{ca}C + \alpha_{da}D + \varepsilon_A, & B &= \alpha_{ab}A + \alpha_{eb}E + \varepsilon_B, \\ C &= \alpha_{bc}B + \varepsilon_C, & D &= \varepsilon_D, \\ E &= \varepsilon_E, & F &= \alpha_{cf}C + \varepsilon_F. \end{aligned}$$

All coefficients are nonzero and $\alpha_{ab}\alpha_{bc}\alpha_{ca} \neq 1$.

- (a) Construct M^{acy} , using [proposition 6.12](#).
- (b) Draw $G^+(M)$, $G^+(M^{\text{acy}})$, $G(M)$ and $G(M^{\text{acy}})$.
- (c) Compute the solution functions $g, \tilde{g}$ and prove $P_M(X_V) = P_{M^{\text{acy}}}(X_V)$.
- (d) Give $T \subseteq V$ and prove that the intervention kernels on $V \setminus T$ differ, so the models are not interventionally equivalent.
- (e) Verify each equivalence by active walks or direct separation arguments:

$$\begin{aligned} D \perp_{G(M)}^\sigma F \mid B &\iff D \perp_{G(M^{\text{acy}})}^d F \mid B, \\ D \perp_{G(M)}^\sigma E \mid A &\iff D \perp_{G(M^{\text{acy}})}^d E \mid A, \\ D \perp_{G(M)}^\sigma E &\iff D \perp_{G(M^{\text{acy}})}^d E. \end{aligned}$$

For the 2024 version, prove (c) and (e) without invoking a theorem that immediately gives the result.

SOLUTION. Let $J = \emptyset$, $V = \{A, B, C, D, E, F\}$, $W = \{A', B', C', D', E', F'\}$, and $\mathcal{X} = \mathbb{R}^{12}$. Let P be the product of independent Gaussian error laws $\varepsilon_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, with positive variances. The equations are

$$\begin{aligned} A &= \alpha_{ca}C + \alpha_{da}D + \varepsilon_A, & B &= \alpha_{ab}A + \alpha_{eb}E + \varepsilon_B, \\ C &= \alpha_{bc}B + \varepsilon_C, & D &= \varepsilon_D, \\ E &= \varepsilon_E, & F &= \alpha_{cf}C + \varepsilon_F. \end{aligned}$$

All coefficients are nonzero and $\Delta = 1 - \alpha_{ab}\alpha_{bc}\alpha_{ca} \neq 0$. The strongly connected components are $\{A, B, C\}$, $\{D\}$, $\{E\}$, and $\{F\}$. By [proposition 6.12](#), define

$$H = \frac{1}{\Delta} \begin{pmatrix} 1 & \alpha_{bc}\alpha_{ca} & \alpha_{ca} \\ \alpha_{ab} & 1 & \alpha_{ab}\alpha_{ca} \\ \alpha_{ab}\alpha_{bc} & \alpha_{bc} & 1 \end{pmatrix}.$$

The acyclification preserves $J, V, W, \mathcal{X}, P$ and replaces the cyclic mechanisms by

$$\begin{pmatrix} \tilde{f}_A \\ \tilde{f}_B \\ \tilde{f}_C \end{pmatrix} = H \begin{pmatrix} \alpha_{da}D + \varepsilon_A \\ \alpha_{eb}E + \varepsilon_B \\ \varepsilon_C \end{pmatrix},$$

while $\tilde{f}_D = \varepsilon_D$, $\tilde{f}_E = \varepsilon_E$, and $\tilde{f}_F = \alpha_{cf}C + \varepsilon_F$.

The original augmented graph adds $i' \rightarrow i$ to the edges

$$A \rightarrow B, \quad B \rightarrow C, \quad C \rightarrow A, \quad D \rightarrow A, \quad E \rightarrow B, \quad C \rightarrow F.$$

In the acyclified augmented graph, D, E and each of A', B', C' point to every node in $\{A, B, C\}$, and $D' \rightarrow D, E' \rightarrow E, F' \rightarrow F, C \rightarrow F$ remain. Its causal graph has

$$D, E \longrightarrow A, B, C, \quad C \longrightarrow F, \quad A \leftrightarrow B, \quad A \leftrightarrow C, \quad B \leftrightarrow C.$$

The four graphs are shown in [figures 6.5 to 6.8](#).

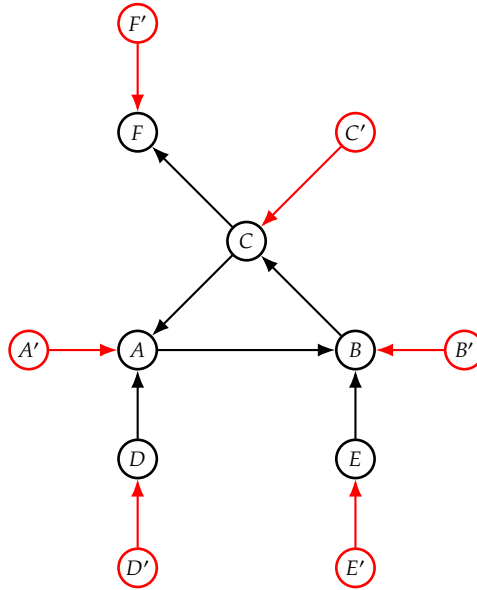


FIGURE 6.5. The augmented graph $G^+(M)$ of the cyclic SCM.

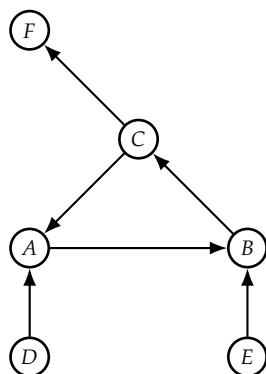
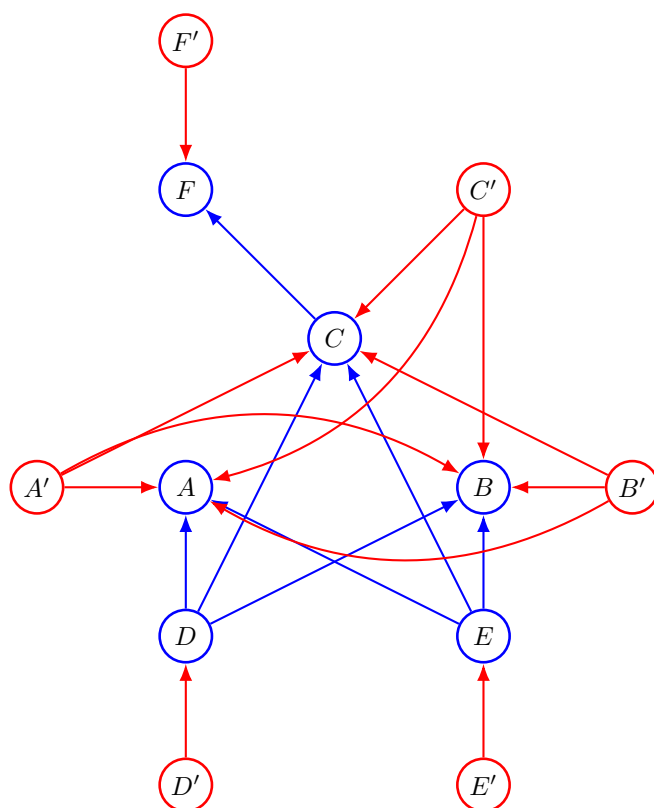
FIGURE 6.6. The causal graph $G(M)$ of the cyclic SCM.

FIGURE 6.7. The augmented graph of the acyclified SCM.

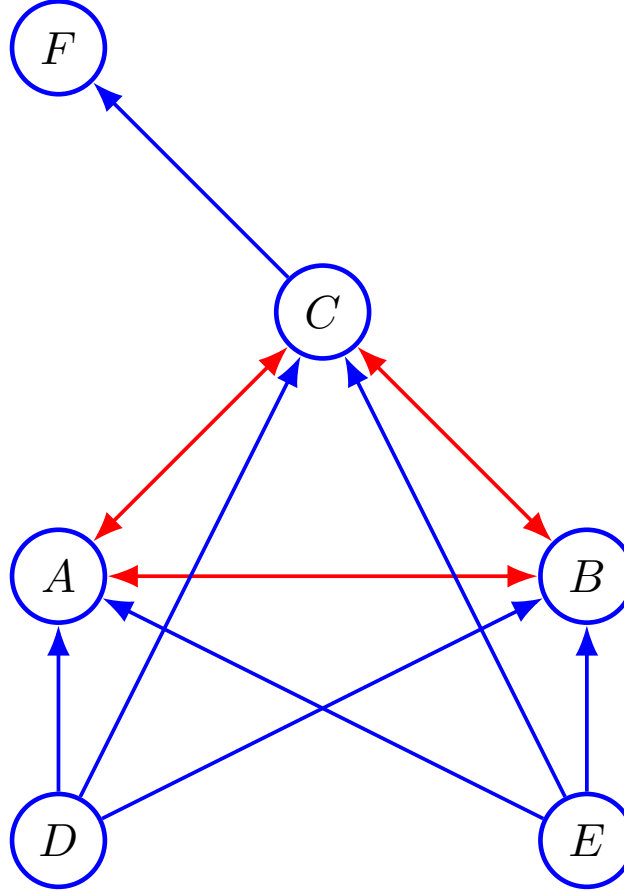


FIGURE 6.8. The causal graph of the acyclified SCM.

Solving either model gives the same function:

$$\begin{pmatrix} g_A \\ g_B \\ g_C \end{pmatrix} = H \begin{pmatrix} \alpha_{da} \varepsilon_D + \varepsilon_A \\ \alpha_{eb} \varepsilon_E + \varepsilon_B \\ \varepsilon_C \end{pmatrix}, \quad g_D = \varepsilon_D, \quad g_E = \varepsilon_E, \quad g_F = \alpha_{cf} g_C + \varepsilon_F.$$

Thus $g = \tilde{g}$ and $P_M(X_V) = g_*P = \tilde{g}_*P = P_{M^{\text{acy}}}(X_V)$. Under $\text{do}(B = b)$, however, the original model gives

$$C = \alpha_{bc}b + \varepsilon_C, \quad P_M(C \mid \text{do}(B = b)) = \mathcal{N}(\alpha_{bc}b + \mu_C, \sigma_C^2).$$

In the acyclified model, C keeps its solved mechanism and its distribution is independent of b . Since $\alpha_{bc} \neq 0$, the intervention kernels cannot agree for every b .

The relevant separation equivalences follow directly from active paths:

- $D \not\perp_{G(M)}^\sigma F \mid B$: the path $D \rightarrow A \rightarrow B \rightarrow C \rightarrow F$ is σ -active because the conditioned node B points only to C within its strongly connected component. In the acyclified graph, $D \rightarrow C \rightarrow F$ is d -active given B .
- $D \not\perp_{G(M)}^\sigma E \mid A$: the path $D \rightarrow A \leftarrow C \leftarrow B \leftarrow E$ is active because the collider A is conditioned on and B, C are unconditioned. In the acyclified graph, $D \rightarrow A \leftarrow E$ is d -active given A .
- $D \perp_{G(M)}^\sigma E$ and $D \perp_{G(M^{\text{acy}})}^d E$: every walk from D to E has a collider. Indeed, both endpoints have only outgoing edges toward the other nodes; a walk beginning with a forward directed edge and ending with a backward directed edge must encounter an arrowhead-to-arrowhead transition. With no conditioning, such a collider is blocked.

□

Causal Domain Adaptation with SCMs

SECTION 7.1

| Do-Calculus for Simple SCMs

Counterfactuals with SCMs

SECTION 8.1

| Bounding Counterfactual Probabilities

Estimating counterfactual probabilities from data can be impossible without relying on very strong modelling assumptions. We can sometimes derive a *bound* on counterfactual probabilities, and one way to derive such bounds on counterfactuals exploits the response function parameterisation.

As a motivation, consider a patient who participated in a randomised controlled trial. The patient was assigned to the placebo group and was not cured. Their physician asks: would the patient have been cured if assigned to the treatment group?

PROPOSITION 8.1.

For a simple SCM M with a single binary exogenous input variable $X \in \{0, 1\}$ and a binary endogenous variable $Y \in \{0, 1\}$ (and perhaps additional endogenous variables, and with an arbitrary number of exogenous random variables taking values in arbitrary standard measurable spaces), the counterfactual probability

$$P_{M^{twin}}\left((Y')^{do(x'=1)} = 1 \mid Y^{do(x=0)} = 0\right)$$

(with $Y^{do(x=0)}$ the factual outcome, and $(Y')^{do(x'=1)}$ the counterfactual outcome) is bounded by

$$\frac{q_{0|0} - \min(q_{0|0}, q_{0|1})}{q_{0|0}} \leq P_{M^{twin}}\left((Y')^{do(x'=1)} = 1 \mid Y^{do(x=0)} = 0\right) \leq \frac{\min(q_{0|0}, q_{1|1})}{q_{0|0}},$$

where $q_{y|x} := P_M(Y = y \mid \text{do}(X = x))$ and $q_{0|0} > 0$.

Proof.

□

EXAMPLE 8.2.

Exercises

EXERCISE 8.1 (Natural Bounds on Causal Effects). Let $J = \emptyset$, $V = \{X, Y\}$, $W = \{Z, U_X, U_Y\}$, with $\mathcal{X}_V = \{0, 1\}^2$, $\mathcal{X}_W = \{0, 1\} \times [0, 1]^2$, independent $Z \sim \text{Bernoulli}(1/3)$ and $U_X, U_Y \sim \mathcal{U}[0, 1]$, and mechanisms

$$X = \mathbb{1}_{\{U_X < (1+Z)/3\}}, \quad Y = \mathbb{1}_{\{U_Y < (1+X+Z)/4\}}.$$

- Find $P(X | Z)$ and $P(Y | X, Z)$ and compare with [exercise 4.4](#).
- Compute natural bounds for $\mathbb{E}[Y | \text{do}(X = 0)]$.
- Compute natural bounds for $\mathbb{E}[Y | \text{do}(X = 1)]$.
- Check the causal means computed in [exercise 4.4](#) against these bounds, as in Theorem 7.5.2.
- Bound the average causal effect and compare with its exact value in [exercise 4.4](#).

EXERCISE 8.2 (The Coin Machine and Counterfactuals). The coin machine in [figure 8.1](#) has a fair output for either input orientation: $P(Y = y | \text{do}(X = x)) = 1/2$, $x, y \in \{0, 1\}$. Here X is an exogenous input and Y endogenous. The 2024 version uses the equivalent coding $\bar{X} = 2X - 1$, $\bar{Y} = 2Y - 1$ in $\{-1, 1\}$.

- Formally construct two SCMs with this kernel but different counterfactual behaviour, and prove that their observational kernels agree. Consider one mechanism ignoring the input and another using it.
- Prove observational and interventional equivalence, but failure of counterfactual equivalence with respect to $\{Y\}$.
- Use [proposition 8.1](#) to bound $P(Y'(1) = 1 | Y(0) = 0)$.
- Let the response function be $f_W(x) = w_x$, with $W = (w_0, w_1) \in \{0, 1\}^2$ and

$$\rho_{00} = \rho_{01} = \alpha/2, \quad \rho_{10} = \rho_{11} = (1 - \alpha)/2, \quad 0 < \alpha < 1.$$

Compute all intervention probabilities and check whether they equal $1/2$. Derive bounds for $P(Y'(1) = 1 | Y(0) = 0)$ for $\alpha < 1/2$ and $\alpha > 1/2$, and analyse the limit $\alpha \rightarrow 1$.

SOLUTION. Use binary coding $X, Y \in \{0, 1\}$ for the coin machine in [figure 8.1](#). Take $J = \{X\}$, $V = \{Y\}$, $W = \{W\}$, $\mathcal{X} = \{0, 1\}^3$, and $W \sim \text{Bernoulli}(1/2)$. Define two SCMs by

$$f(x, y, w) = w, \quad \tilde{f}(x, y, w) = x \oplus w.$$

Their solution functions are $g(x, w) = w$ and $\tilde{g}(x, w) = x \oplus w$, and both satisfy

$$P(Y = y | \text{do}(X = x)) = \frac{1}{2}, \quad x, y \in \{0, 1\}.$$

For each fixed x , both W and $x \oplus W$ are uniformly distributed. Thus the observational kernels coincide. Intervening on Y fixes the output in both models, so the models are also interventionally equivalent with respect to $\{Y\}$.

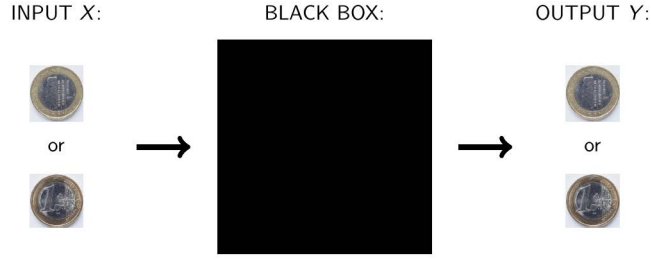


FIGURE 8.1. A coin machine with a fair output for either input orientation.

The twin models share W between the factual and counterfactual worlds, with input nodes $\{X, X'\}$, output nodes $\{Y, Y'\}$, binary product space $\{0, 1\}^5$, and mechanisms

$$f^{\text{twin}} = (W, W), \quad \tilde{f}^{\text{twin}} = (X \oplus W, X' \oplus W).$$

For $x = 0, x' = 1$, their joint outcome distributions are

y	y'	$P_{M^{\text{twin}}}(y, y' \mid 0, 1)$	$P_{\tilde{M}^{\text{twin}}}(y, y' \mid 0, 1)$
0	0	1/2	0
1	1	1/2	0
0	1	0	1/2
1	0	0	1/2

and hence

$$P_{M^{\text{twin}}}(Y'(1) = 1 \mid Y(0) = 0) = 0, \quad P_{\tilde{M}^{\text{twin}}}(Y'(1) = 1 \mid Y(0) = 0) = 1.$$

The models are therefore not counterfactually equivalent. Since $q_{0|0} = q_{0|1} = q_{1|1} = 1/2$, [proposition 8.1](#) gives the sharp bounds $[0, 1]$ from the intervention kernels alone.

Let $W = (Y(0), Y(1)) \in \{0, 1\}^2$, with response functions

x	$f_{0,0}(x)$	$f_{0,1}(x)$	$f_{1,0}(x)$	$f_{1,1}(x)$
0	0	0	1	1
1	0	1	0	1

and probabilities

$$\rho_{0,0} = \rho_{0,1} = \frac{\alpha}{2}, \quad \rho_{1,0} = \rho_{1,1} = \frac{1-\alpha}{2}, \quad 0 < \alpha < 1.$$

The intervention probabilities are

$$q_{0|0} = \alpha, \quad q_{1|0} = 1 - \alpha, \quad q_{0|1} = q_{1|1} = \frac{1}{2}.$$

Thus all four probabilities equal $1/2$ only when $\alpha = 1/2$. Writing $r = P(Y'(1) = 1 \mid Y(0) = 0)$, [proposition 8.1](#) gives

$$\begin{cases} 0 \leq r \leq 1, & 0 < \alpha \leq 1/2, \\ 1 - \frac{1}{2\alpha} \leq r \leq \frac{1}{2\alpha}, & 1/2 < \alpha < 1. \end{cases}$$

Both endpoints converge to $1/2$ as $\alpha \rightarrow 1$. For the specified response distribution itself,

$$r = \frac{\rho_{0,1}}{\rho_{0,0} + \rho_{0,1}} = \frac{1}{2}.$$

□

EXERCISE 8.3 (Counterfactual Bounds with Confounding). Consider $J = \emptyset$, $V = \{X, Y\}$, $W = \{U\}$, $\mathcal{X} = \{0, 1\}^3$, $U \sim \text{Bernoulli}(p_U)$ with $0 < p_U < 1$, and mechanisms $X = f_X(U)$ and $Y = f_Y(X, U)$.

- (a) Give a response-variable encoding M_ρ with $U \in \{0, 1\}^3$ and distribution $\rho \in \Delta_7$, where $\Delta_7 = \{\rho \in [0, 1]^8 : \sum_i \rho_i = 1\}$. For the 2024 version, justify counterfactual equivalence.
- (b) Express $P(X = x, Y = y)$ and $P(Y = y \mid \text{do}(X = x))$ in terms of ρ for each $(x, y) \in \{0, 1\}^2$.
- (c) Express $P_{M_\rho^{\text{twin}}}(U \mid X = 0, Y = 0)$ in terms of ρ .
- (d) Express $P_{M_\rho^{\text{twin}}}(Y'(1) = 1 \mid Y = 0, X = 0)$ in terms of ρ .
- (e) For any observed probabilities p_{xy} with $p_{00} > 0$, show that parameters fitting the observations can make this counterfactual probability take any value in $(0, 1)$. Conclude that observational data alone give no informative bound.

Causal Relations and Confounding

| Exercises

EXERCISE 9.1 (Reichenbach's Principle). Let M be a simple SCM with observed endogenous X, Y , possibly other latent variables, and no exogenous input nodes. Prove that $X \not\perp_{P_M} Y$ implies at least one of $X \rightarrow Y$, $Y \rightarrow X$, or $X \leftrightarrow Y$ in the causal graph after marginalising all endogenous nodes except X, Y .

EXERCISE 9.2 (Fairness in Berkeley Admissions). In 1973, Berkeley received 12763 graduate applications: 8442 male applicants had an aggregate admission rate of about 44.2%, and 4321 female applicants about 34.6%. Within departments, admission rates did not differ significantly by sex; women applied more often to departments with lower admission rates. Let S, A, D denote sex, admission and department choice.

- (a) Specify an SCM for the admission process and draw its graph.
- (b) Which causal effect is regarded as unfair, and whose absence as fair?
- (c) Observational fairness means $P_M(A \mid S, D = d) = P_M(A \mid D = d)$ for every d . Define interventional, counterfactual and graphical fairness using the corresponding causal-relation notions.
- (d) Derive implications between these three causal fairness notions using Proposition 9.2.16. Relate them to observational fairness and derive corresponding implications for unfairness.
- (e) Assume $P_M(S, D) > 0$ and no confounding between department choice and admission. Given that the four fairness notions then agree and yield the same set of observational distributions, explain how to test each notion statistically. *Hint. Test observational fairness.*
- (f) Explain why rejecting observational fairness implies graphical unfairness under no confounding between A, D , and why this implication can fail with confounding.
- (g) With confounding between A, D , construct a model that is interventionally fair but not graphically fair.

Causal Discovery & Estimation with SCMs

| Exercises

EXERCISE 10.1 (Y-Structures). Prove the Y-structure proposition in Proposition 10.4.1 using the global Markov property and faithfulness. Show that the only cyclic or acyclic graphs compatible with the observed conditional independences are those in the proposition's figure (Figure 26 in 2024; Figure 28 in 2025), following the approach used for local causal discovery.

EXERCISE 10.2 (Causal Discovery Programming). Complete Exercises 3–6 of the programming notebook linked in [exercise 11.1](#).

Independence Testing

| Exercises

EXERCISE 11.1 (Conditional Independence Testing). Complete Exercises 1 and 2 of the programming notebook:

https://colab.research.google.com/drive/1uB64Ir9vHT9x_2qM0b3CMHse93Gl_Fdm?usp=sharing

Read its initial instructions carefully.

EXERCISE 11.2 (Causal Discovery and Independence Testing). Complete the programming exercise in the notebook linked in *exercise 11.1*. Read its initial instructions carefully.

Final Exams

Semester 2025–2026

EXERCISE A.1 (Do-Calculus).

EXERCISE A.2 (Conditional Independence).

EXERCISE A.3 (Price, Supply, Demand).

EXERCISE A.4 (Tariffs).

EXERCISE A.5 (Local Markov Property).

Semester 2024–2025

EXERCISE A.6 (Do-Calculus). Consider the observational graph of a causal Bayesian network with latent variables in Figure 1. We are interested in estimating the causal effect of X and W on Y , that is, the interventional distribution $P(Y \mid \text{do}(X, W))$. However, we are only able to perform interventional experiments where we intervene on W . In other words, we are only able to produce samples from the interventional distribution: $P(X, Y, Z \mid \text{do}(W))$. As a first step, use the do-calculus to prove the following *a.s.* equalities:

- (i) $P(Z \mid \text{do}(X, W)) = P(Z \mid X, \text{do}(W))$.
- (ii) $P(Y \mid Z, \text{do}(X, W)) = P(Y \mid \text{do}(X, Z, W))$.
- (iii) $P(Y \mid \text{do}(X, Z, W)) = P(Y \mid \text{do}(Z, W))$.
- (iv) $P(Y \mid \text{do}(X, Z, W)) = P(Y \mid \text{do}(X, W))$.
- (v) $P(X \mid \text{do}(Z, W)) = P(X \mid Z, \text{do}(W))$.

To avoid measure-theoretic subtleties, you are allowed to assume (without further justification) that all occurring Markov kernels are discrete and have strictly positive mass functions. For each point, provide all details of your arguments. When you use d -separation statements clearly indicate in which graph this should hold and draw the graph. Also argue why the claimed d -separation statements hold by listing all relevant paths that need checking and indicate why they are d -blocked. Now we want to use (some/all) of the above found relations to solve the initial problem.

- (vi) Derive an expression for the Markov kernel $P(Y \mid \text{do}(X, W))$ in terms of (products and compositions of marginals and conditionals of) $P(X, Y, Z \mid \text{do}(W))$.

Hint. Make use of the chain rule as follows:

$$P(Y \mid \text{do}(X, W)) = P(Y \mid Z, \text{do}(X, W)) \circ P(Z \mid \text{do}(X, W)),$$

$$P(Y \mid \text{do}(Z, W)) = P(Y \mid X, \text{do}(Z, W)) \circ P(X \mid \text{do}(Z, W)).$$

EXERCISE A.7 (Reichenbach's Principle). Hans Reichenbach (1891-1953), a German physicist, philosopher and logician, introduced a principle named after him in his book *The Direction of Time*. Reichenbach's *Principle of Common Cause* states that if two events are dependent, then one must cause the other, or the events must have a common cause.

One could criticise this principle by arguing that it is not exhaustive: one can think of other reasons for the observed dependence apart from the three mentioned in *Reichenbach's principle*. We will take a closer look at the situation using the framework of simple SCMs.

- (i) Let M be a simple SCM with two observed endogenous variables X, Y , an arbitrary number and type of exogenous random variables, but no exogenous input variables. Prove that $X \not\perp\!\!\!\perp Y$ implies at least one of the following:
- X causes Y according to $G(M)$.
 - Y causes X according to $G(M)$.
 - X and Y are confounded according to $G(M)$.
- (ii) Use this result to prove that the same statement holds for simple SCMs with endogenous variables $V \supseteq \{X, Y\}$, no restrictions on the exogenous random variables, and without exogenous input variables.
- (iii) Give another possible explanation of an observed dependence between X and Y , which is not mentioned in *Reichenbach's principle*.

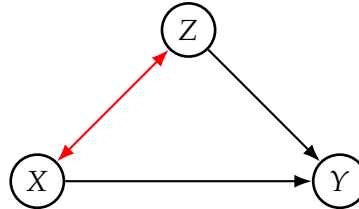
EXERCISE A.8 (The Risk of Dying from COVID-19). In this exercise we investigate the influence of age and country on the risk of dying from a COVID-19 infection. Figure 2 displays data from the beginning of the COVID-19 pandemic in 2020, which started with an outbreak in China followed a few weeks later by an outbreak in Italy. Figure 2 (left) shows COVID-19 CFRs (case fatality rates; that is, the proportion of confirmed COVID-19 cases that end with the death of the patient); for both countries, the CFRs are displayed for different age groups separately, and in addition the total CFR across all age groups combined is shown. Figure 2 (right) shows for both countries how the confirmed COVID-19 cases are distributed over the age groups.

Interestingly, while for all age groups, CFRs in Italy are lower than in China, the total CFR in Italy is higher than in China. This is an example of a phenomenon known as *Simpson's paradox*.

We decide to model the situation in the following way. For each confirmed case (a person who tested positive within a certain time frame and was reported by health officials of a certain country), we record the country, age group and fatality outcome. Thus, we associate to each confirmed case a tuple of three random variables:

Name	Symbol	Meaning
Country	X	Country in which the case is reported
Age	Z	Age group to which the person belongs
Fatality	Y	Binary indicator of whether the person has died by the time of reporting

We assume that the data-generating process can be modelled with a structural causal model with the following causal graph:



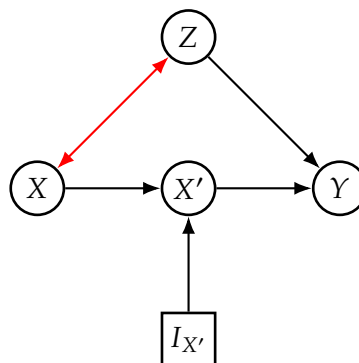
- Under what circumstances is it realistic to assume that the edges $Y \rightarrow X$ and $Y \rightarrow Z$ are both absent from the causal graph?
- Explain why it is sensible to assume that $X \rightarrow Z$ and $Z \rightarrow X$ are both absent from the causal graph.
- Prove that if the edge $X \leftrightarrow Z$ were absent from the causal graph, then country and age would be independent.
- Give the technical term for what the edge $X \leftrightarrow Z$ represents. Explain with the help of Figure 2 whether you think that the bi-directed edge $X \leftrightarrow Z$ should indeed be part of the model, or whether it could be safely omitted.
- One interesting quantity to consider is the average total causal effect of country on fatality, defined as:

$$\rho := \mathbb{E}[Y \mid \text{do}(X = 1)] - \mathbb{E}[Y \mid \text{do}(X = 0)].$$

Give an identity that expresses this quantity purely in terms of the observed distribution $P(X, Y, Z)$.

Hint. You may make use of the backdoor criterion.

Another question of interest is: for the Chinese case demographic, would the Italian healthcare system have been better? More specifically: what is the change in the risk of dying from COVID-19 when a randomly selected positive case from China is forced to move to Italy? We can model this by adding a copy X' of X (interpreted as the value of “country” measured later in time) and an intervention variable $I_{X'}$. The causal graph then becomes:



We assume that if we don't intervene on X' , its value is just $X' = X$. In this setting, we can consider the quantity:¹

$$\tau := \sum_z (\mathbb{E}[Y \mid Z = z, X = 1] - \mathbb{E}[Y \mid Z = z, X = 0])P(Z = z \mid X = 0).$$

(vi) Prove that τ can be expressed in terms of the observed distribution $P(X, Y, Z)$ as:

$$\tau = \sum_z (\mathbb{E}[Y \mid Z = z, do(X' = 1)] - \mathbb{E}[Y \mid Z = z, do(X' = 0)])P(Z = z \mid X = 0) \quad a.s.$$

To simplify matters, you may assume that all relevant Markov kernels are strictly positive.

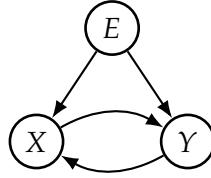
(vii) How could you explain *Simpson's paradox* in this case?

EXERCISE A.9 (Simple SCM with Binary Variables). (i) Let M be an SCM with two endogenous variables X, Y , no exogenous input variables, and no exogenous random variables, and graph $G^+(M)$:



Prove that M cannot be simple if X and Y are both binary (that is, if they take values in the space $\{0, 1\}$).

(ii) Give an example of a simple SCM $\tilde{M}$ with two binary endogenous variables X, Y , no exogenous input variables, and one exogenous random variable E that has graph $G^+(\tilde{M})$:



Prove that it is indeed simple.

EXERCISE A.10 (Separoid Rules). In the following let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}, \mathcal{W}, \mathcal{U}$ and $\mathcal{T}$ be standard measurable spaces and

$$K(W \mid T) : \mathcal{T} \rightarrow \mathcal{W}$$

be a Markov kernel. Furthermore, we consider corresponding conditional random variables:

$$X : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{X},$$

$$Y : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{Y},$$

$$Z : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{Z},$$

and so on, and let $T : \mathcal{W} \times \mathcal{T} \rightarrow \mathcal{T}$ be the canonical projection onto the second argument, and let $*$ be the constant map. We write $X \perp\!\!\!\perp Y \mid Z$ for the conditional independence *w.r.t.* $K(W \mid T)$. Prove the following implications.²

¹ This quantity is also sometimes called the “effect of treatment on the treated”.

² We write $X \preceq Y$ if and only if there exists a measurable map $f : \mathcal{Y} \rightarrow \mathcal{X}$ such that $X = f \circ Y$. We also write: $X \approx Y$ if and only if $X \preceq Y$ and $Y \preceq X$.

- (i) $T \lesssim Z \implies X \perp\!\!\!\perp T \mid Z$.
- (ii) $U \lesssim (Y, Z)$ and $X \perp\!\!\!\perp Y \mid Z \implies X \perp\!\!\!\perp U, Y \mid Z$.
- (iii) $U \approx Z$ and $X \perp\!\!\!\perp Y \mid Z \implies X \perp\!\!\!\perp Y \mid U$.

Hint. All those rules can be proven only with help of the standard (asymmetric) separoid rules and the following (trivial) observation:

$$U \lesssim Z \implies U \lesssim (Y, Z).$$

You are also allowed to use rule (ii) to prove rule (i) above.

Semester 2023–2024

EXERCISE A.11 (Graphs).

EXERCISE A.12 (Marginalisation).

EXERCISE A.13 (Do-Calculus).

EXERCISE A.14 (Selection Bias).

EXERCISE A.15 (Instrumental Variables).

References

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.
- [2] Patrick Forré and Joris M. Mooij. A mathematical introduction to causality: Lecture notes. University of Amsterdam, 2025.